



**PHYSICAL SCIENCES
GRADE 12**

PAPER 1

(PHYSICS FOCUS)

**ELECTRIC CIRCUITS WITH INTERNAL
RESISTANCE**

4,5 & 6 JULY 2025

LESSONS 1,2 AND 3

Concepts Check - PAPER 1 (PHYSICS)

CONTENT	SUB-TOPIC	ESTIMATED MARKS	TOTAL MARKS
Mechanics	Newton's laws and applications	± 18	65
	Vertical projectile motion	± 17	
	Conservation of linear momentum	± 14	
	Work , energy and power	± 17	
Waves, Sound and Light	Doppler effect: Blue shift & Red shift	± 15	15
Electricity and Magnetism	Electrostatics (Coulomb's law ; Electric fields)	± 20	55
	Electric circuits	± 18	
	Electrodynamics(Electric machines)	± 16	
Matter and Materials	Photo electric effect Spectrum (Absorption & Emission)	± 15	15
	TOTAL		150

Waves Sound & Light – Photoelectric Effect (P1)	
$E = hf = W_0 + E_{max}$ $W_0 = hf_0$ $E_{k(max)} = \frac{1}{2}mv_{max}^2$	E = energy of photon (J). h = Planck's constant ($h = 6,63 \times 10^{-34}$ J·s) W_0 = work function of a metal (J) f_0 = cut-off /threshold frequency (Hz) m = mass of a photo-electron (kg) $(m_e = 9,11 \times 10^{-31}$ kg) v = speed of emitted photoelectrons ($m \cdot s^{-1}$)
Electricity & Magnetism - Electrostatics(P1)	
$F = \frac{kQ_1Q_2}{r^2}$ $E = \frac{kQ}{r^2}$ $E = \frac{F}{q}$ $V = \frac{W}{q}$ $n = \frac{Q}{q_e}$	F = electrostatic force (N) b/w 2 charges r m apart E = strength of electric field due to charge Q (at distance r from Q) $(k = \text{Coulomb's constant} = 9,0 \times 10^9 \text{ N} \cdot \text{m}^2 \cdot \text{C}^{-2})$ Q = charge (coulomb, C) $E = F/q$ — electric field strength at test charge q V = potential difference (volts, V) W = work done (joule, J) n = number of electrons e or q_e = charge of an electron (C) ($= -1,6 \times 10^{-19}$ C)
Electricity & Magnetism - Electric Circuits (P1)	
$R_s = R_1 + R_2 + \dots$ $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$	R = resistance (Ohm, Ω) s / p = series / parallel R_s, R_p = total resistance of a combination of resistors in series / in parallel
Electricity & Magnetism - Electric Circuits cont...	
$\varepsilon = I(R + r)$ $Q = I\Delta t$ $W = Vq$ $W = I^2R\Delta t$ $P = VI$	$V = IR$ $W = VI\Delta t$ $W = \frac{V^2\Delta t}{R}$ $P = \frac{W}{\Delta t}$ $P = \frac{V^2}{R}$ $P = I^2R$
$\varepsilon = \text{emf (volts, V)}$ $r = \text{internal resistance of battery } (\Omega)$ $R = \text{resistance of a resistor/ total external resistance of circuit } (\Omega)$ $I = \text{current (ampere, A)}$ $Q \text{ (or } q) = \text{charge (coulomb, C)}$ $\Delta t = \text{time elapsed (s)}$ $V = \text{potential diff. (volts, V)}$ $W = \text{work done (joule, J)}$ $P = \text{power (watt, W)}$	
Electricity & Magnetism - Alternating Current (P1)	
$I_{rms} = \frac{I_{max}}{\sqrt{2}}$ $V_{rms} = \frac{V_{max}}{\sqrt{2}}$ $P_{ave} = V_{rms} \cdot I_{rms}$ $P_{ave} = I_{rms}^2 \cdot R$ $P_{ave} = \frac{V_{rms}^2}{R}$	I_{rms} = root mean square current (A) V_{rms} = root mean square pot. difference (V) P_{ave} = average power (watt, W) R = resistance of a resistor (ohm, Ω)



(Electric Circuits)

4 JULY 2025

LESSON 1

GR 10-11 CONCEPTS

RECAP

ELECTRIC CIRCUIT - GRADE 10

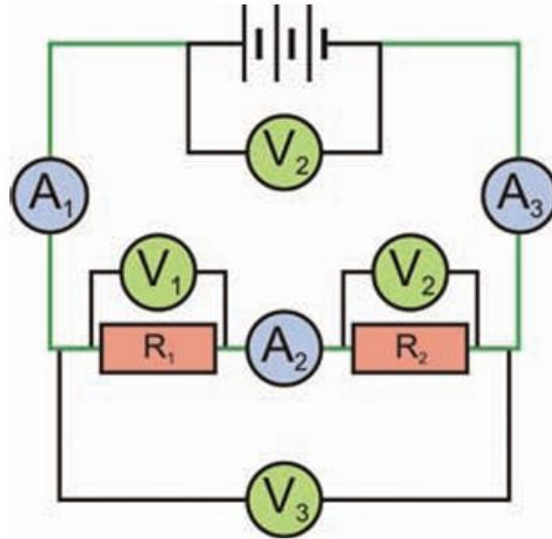
Electric Circuits	Syllabus Requirement
Emf Terminal potential difference (pd)	□ Define potential difference in terms of work done and charge ($V = W/Q$) □ Know that voltage measured across the terminals of a battery when: ◆ no current is flowing through the battery is called the emf. ◆ a current is flowing through the battery is called terminal potential difference (pd). □ Unit: Emf and pd are measure in volts (V)
Current	□ Define current, I (<i>rate of flow of charge</i>) □ Unit - measured in ampere (A), =coulomb per second □ Calculate current flowing ,use equation $I = \frac{Q}{\Delta t}$ □ Indicate the direction of the current in circuit diagrams (conventional)
Measurement of voltage(pd) and current	□ Draw a diagram to show how to correctly connect: ◆ an ammeter to measure the current through a given circuit element ◆ a voltmeter to measure the voltage across a given circuit element

Electric Circuits	Syllabus Requirement
Resistance	□ Define resistance (<i>opposition to the flow of electric current</i>) □ Unit: of resistance; one ohm (Ω)-one volt per ampere. □ Give a microscopic description of resistance - in terms of electrons moving through a conductor colliding with the particles of which the conductor (metal) is made and transferring kinetic energy. □ Explain ◆ And state factors that affect resistance of a substance ◆ Why a battery in a circuit goes flat eventually - refer to the energy transformations that take place in the battery and the resistors in a circuit
Resistors in series	□ Know that: ◆ Current is constant through each resistor in series circuit. ◆ Series circuits are called voltage dividers because the total potential difference is equal to the sum of the potential differences across all the individual components. □ Calculate the equivalent (total) resistance of resistors connected in series using: $R_s = R_1 + R_2 + \dots$

Electric Circuits	Syllabus Requirement
Resistors in parallel	□ Know that: • Voltage is constant across resistors connected in parallel. • A parallel circuit is a current divider - the total current in the circuit is equal to the sum of the branch currents using $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$ • Know that for TWO resistors connected in parallel, the total resistance can be calculated using:$R_p = \frac{\text{product}}{\text{sum}} = \frac{R_1 R_2}{R_1 + R_2}$

ELECTRIC CIRCUIT - GRADE 10

Resistors in series



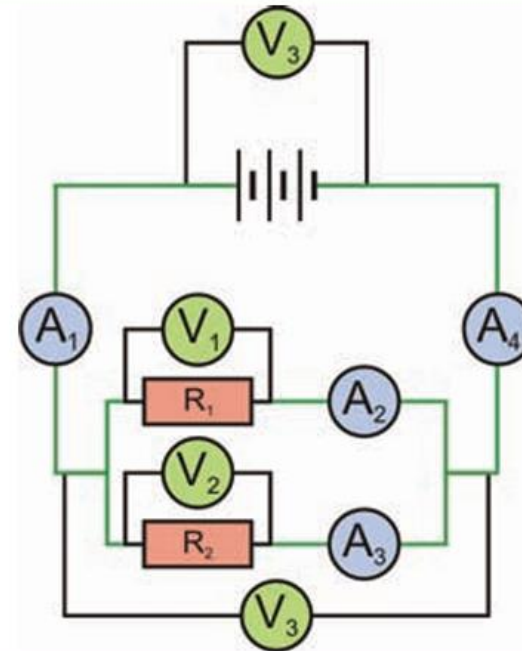
The current (I) is the same anywhere in a series circuit.

$$I_1 = I_2 = I_3$$

Resistors in series are voltage dividers:

$$V_1 + V_2 = V_3$$

Resistors in parallel



$$I_1 = I_4$$

Resistors in parallel are current dividers:

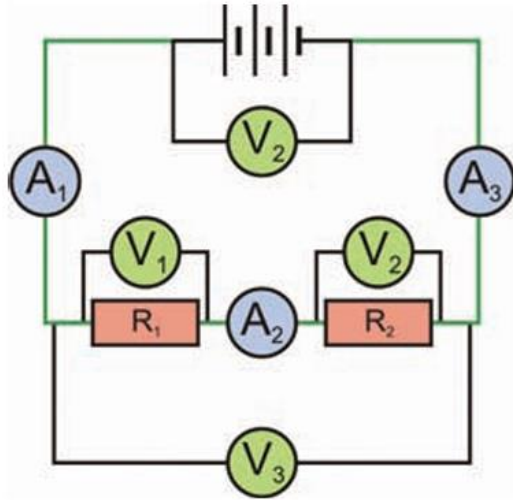
$$I_2 + I_3 = I_1$$

$$I_2 + I_3 = I_4$$

V_1 and V_2 will always have the same reading. Also, if a voltmeter V_3 is connected across the whole parallel branch, it will also read the same as V_1 and V_2 .

$$V_1 = V_2 = V_3$$

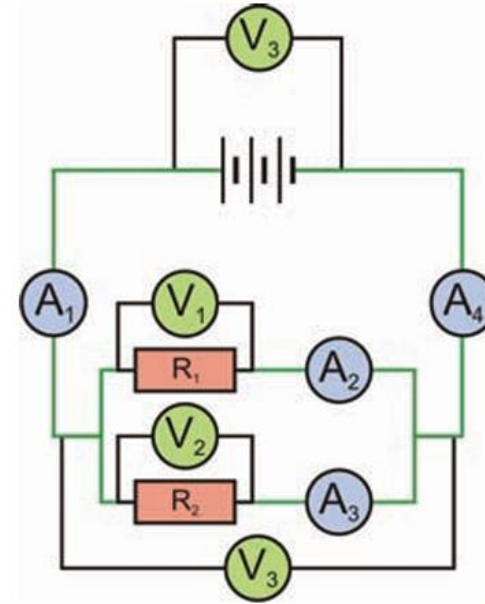
Resistors in series



To calculate the total resistance of any number of resistors connected in series, we use the equation below:

$$R_S = R_1 + R_2 + R_3 + \dots$$

Resistors in parallel



To calculate the total resistance when resistors are connected in parallel, we use the equation below:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \dots$$

OR

For 2 resistances in parallel:

$$R_p = \frac{\text{product}}{\text{sum}} \\ = \frac{R_1 \times R_2}{R_1 + R_2}$$

Electric circuits	Syllabus Requirement
	All concepts covered in grade 10. In addition to grades 10 content , grade 11 learners will be required to do the following:
Relation between current, voltage and resistance (Ohm's Law)	□ Determine the relationship between current, voltage and resistance at constant temperature using a simple circuit. □ State the difference between ohmic and non-ohmic conductors and give an example of each. (light bulb ohmic; Nichrome non-ohmic). □ Calculate/Solve problems using the mathematical expression of Ohm's Law, $R=V/I$, for: ◆ series circuits ◆ parallel circuits
Resistance, equivalent resistance, internal resistance	□ Calculate the equivalent resistance of series and parallel arrangements of resistors (Maximum of FOUR resistors). □ Solve problems involving current, voltage and resistance for circuits containing arrangements of resistors in series and in parallel.
Series, parallel networks	□ Solve circuit problems involving resistors in series with parallel networks of resistors

Power, Energy

- Define:
 - ◆ power (the rate at which electrical energy is converted in an electric circuit and is measured in watts (W))
- Know:
 - ◆ Electrical power dissipated in a device is equal to the product of the potential difference across the device and current flowing through it i.e. $P = IV$
 - ◆ Power can also be given by $P = I^2R$ or $P = \frac{V^2}{R}$
 - ◆ Electrical energy is given by $E = Pt$ and is measured in joules (J)
 - ◆ The kilowatt hour (kWh) refers to use of 1 kilowatt of electricity for 1 hour
- Solve/calculate problems involving:
 - ◆ Circuits, including the concept of power
 - ◆ The concept of electrical energy
- The cost of electricity usage, given the power specifications of appliances used and the duration if the cost of 1 kWh is given

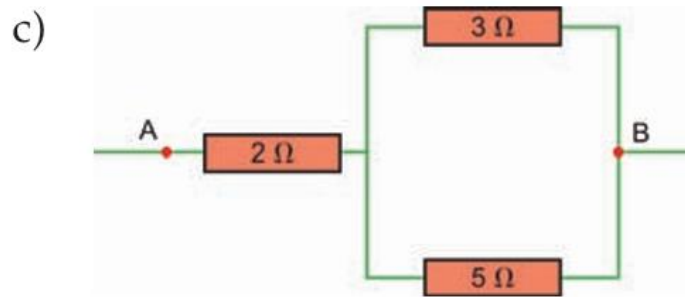
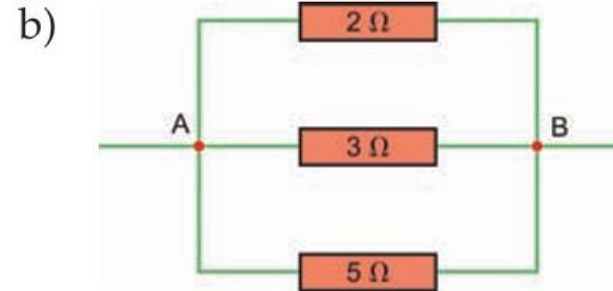
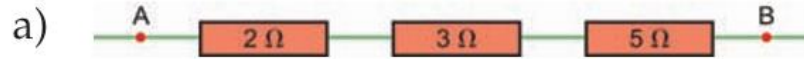
ELECTRIC CIRCUIT - GRADE 12

Electric circuit	Syllabus Requirement
	All content covered in grades 10 and 11. In addition to grades 10 and 11 content, grade 12 learners will be required to do the following:
Resistance, equivalent resistance, internal resistance	□ State that a real battery has internal resistance. □ Explain why there is a difference between the emf and terminal voltage of a battery if the load (external resistance in the circuit) is comparable in size to the battery's internal resistance. □ Solve circuit problems in which the internal resistance of the battery must be considered.
Series, parallel networks	□ Solve circuit problems involving resistors in series with parallel networks of resistors (Maximum of FOUR resistors). □ Calculations - Solve circuit problems involving: ◆ The sum of the voltages across the external circuit plus the voltage across the internal resistance is equal to the emf: $\mathcal{E} = V_{\text{load}} + V_{\text{internal resistance}} \text{ or } \mathcal{E} = IR_{\text{ext}} + Ir$ ◆ current, voltage and resistance for circuits containing arrangements of resistors in series and in parallel □ the internal resistance of the battery, involving series-parallel networks of resistors

ELECTRIC CIRCUIT (GR 10 & 11 - ACTIVITIES)

1.
 - a) Define potential difference.
 - b) Write down an equation which defines potential difference in terms of work done (W) and charge (Q) moved.
 - c) Define emf.
 - d) What is the unit of potential difference and emf?
2. What is the potential difference between two points in a circuit when 80 C of charge transfers 4000 J of energy as it moves between the two points?
3.
 - a) Define electric current.
 - b) State the unit of electric current.
 - c) Write down the equation which defines electric current.
4. The potential difference across a light bulb in an electric circuit is 15 V. How many joules of electrical energy are changed into light and heat energy when there is a current of 2 A in the light bulb for 10 s?
5. An electric toaster, operating at a potential difference of 115 V, uses 34 200 J of energy in 3 minutes. Calculate the current in the heating element of the toaster.

6. Calculate the equivalent (total) resistance between points A and B in each of the following cases:



7. State two ways of increasing the current in an electrical circuit.
8. State Ohm's law.
9. Explain why the resistance of a metal wire increases when it is heated.
10. a) Describe the difference between current in resistors connected in series and current in resistors connected in parallel.
- b) Describe the difference between voltage across resistors connect in series and voltage across resistors connected in parallel.

11. Calculate the unknown values in each of the following circuit diagrams:

a) Refer to Figure 9.1 and calculate the unknown values R_2 , R_3 , V_1 and V_3 .

b) Refer to Figure 9.2 and calculate the unknown values I_1 , I_2 , I_3 , V_1 and V_2 .

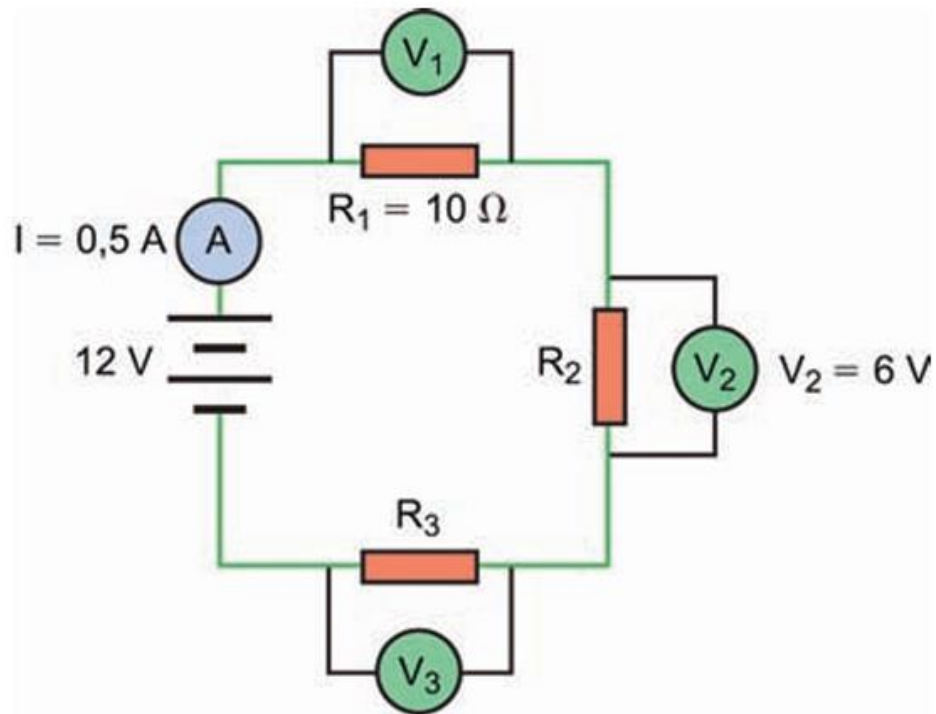


Fig 9.1

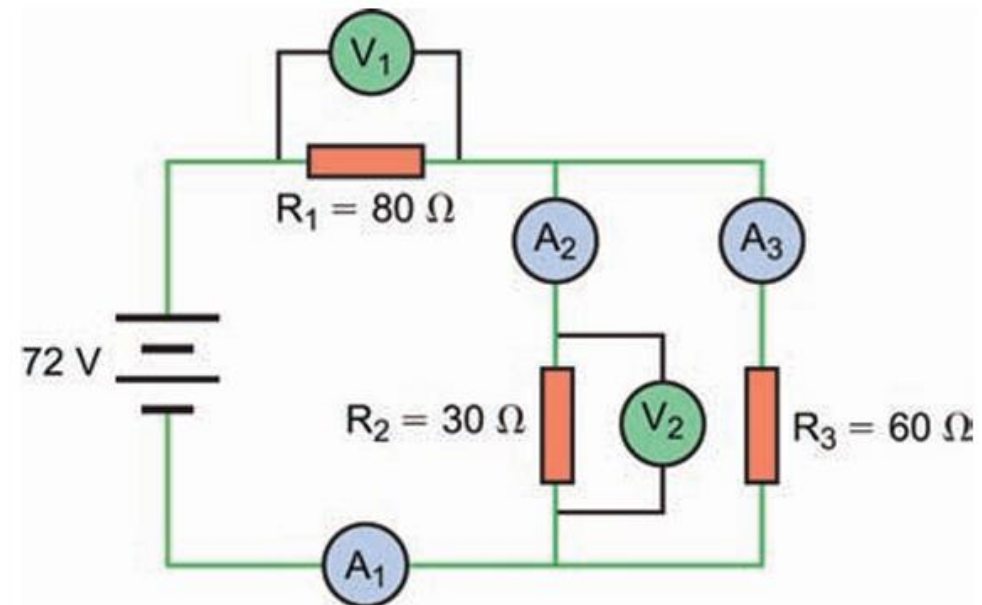
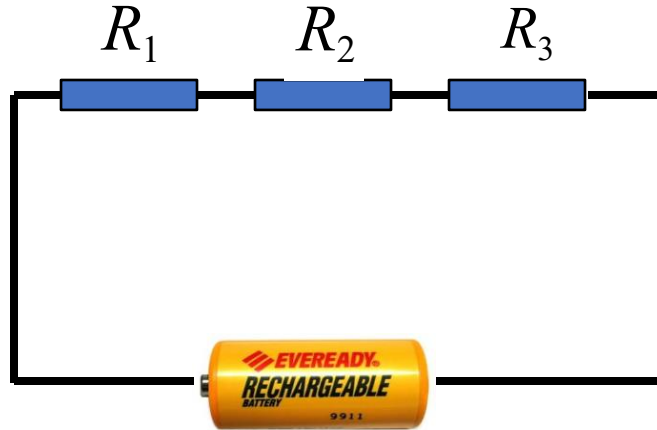


Fig 9.2

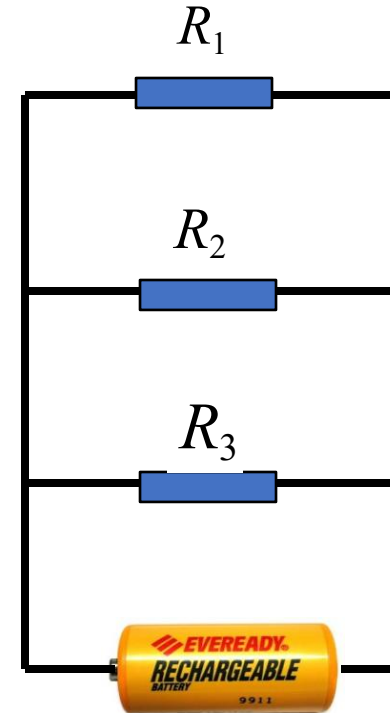
Special concepts about circuits

Series Circuit



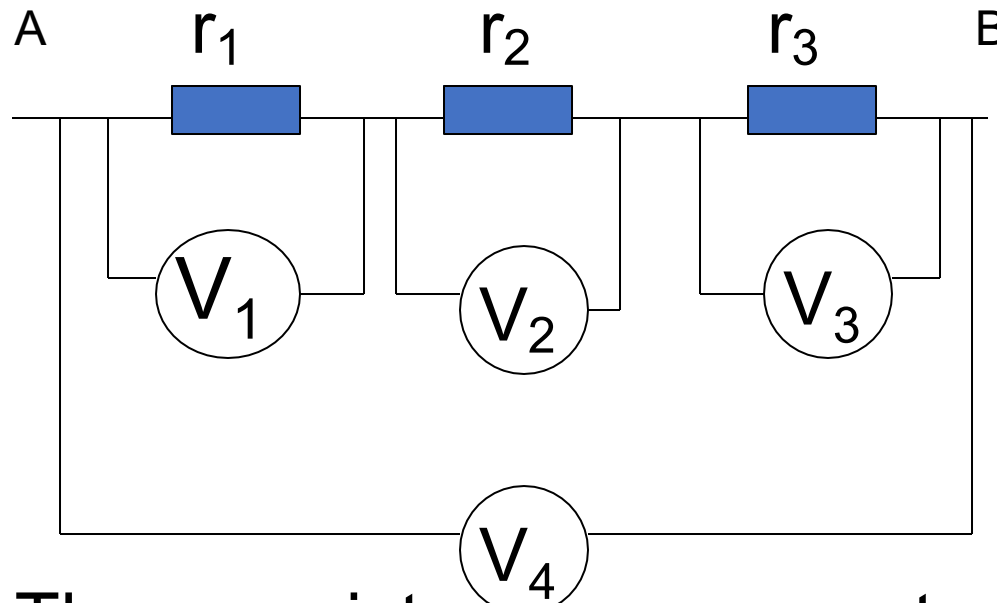
$$R_s = R_1 + R_2 + R_3$$

Parallel Circuit



$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Series connection



 = Resistor

For current to pass from A to B, it must pass through each resistor.

These resistors are connected in series.

$$R_{\text{tot}} = r_1 + r_2 + r_3 \quad \& \quad V_{\text{tot}} = V_4 = V_1 + V_2 + V_3$$

Connecting resistors in series increases resistance and divides the potential difference.

RESISTORS IN SERIES

1. In a series circuit, certain general rules may be stated with regard to quantities of voltage, current, resistance, and power:

❖ "In a series circuit, **voltage** drops add to equal the total."

❖ "In a series circuit, **current** is equal through all components."

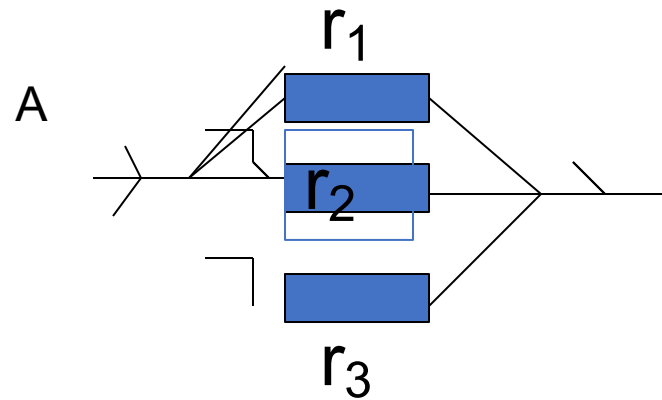
❖ "In a series circuit, **resistances** add to equal the total."

❖ "In a series circuit, **power** dissipations add to equal the total."

RESISTORS IN SERIES CONT...

1. In a series circuit, the **total resistance** increases as more resistors are added.
2. In a series circuit, if one resistor is not working (burns out), the circuit is broken and no current will flow through the circuit.
3. In a series circuit, the ***total resistance*** will **always be greater than** that of the **resistor with the greatest resistance** in the circuit.
4. In a series circuit, transfer of ***power*** will be the ***greatest*** with the ***resistor with the greatest resistance*** according to, $P = I^2R$
5. In a series circuit, the ***resistor with the greatest resistance*** will also have the ***greatest*** voltmeter reading (***potential difference***) according to $V = IR$ and according to $P = VI$ also the ***greatest transfer of power***.

Parallel connection



These resistors are connected in parallel and for current to go from A to B, some will go through each resistor

Connecting resistors in parallel, decreases the effective resistance.

Parallel resistors are called current dividers.

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$I_T = I_1 + I_2 + I_3$$

RESISTORS IN PARALLEL

1. In a parallel circuit, certain general rules may be stated with regard to quantities of voltage, current, resistance, and power.
 - ❖ In a parallel circuit, **voltage** *across all components equals voltage across each component/*
 - ❖ In a parallel circuit, **currents** *add to equal the total."*
 - ❖ In a parallel circuit, **resistance of resistors** *diminish to equal the total."*
 - ❖ In a parallel circuit, **power** *dissipations add to equal the total."*

RESISTORS IN PARALLELE CONT...

1. The equivalent resistance is always less than the resistance of the smallest resistor in the group.
2. In parallel circuit, each device operates independently of the others so that if one is switched off, the others remain on.
3. In parallel circuit, all of the devices operate on the same voltage.
4. The current takes all the paths.
 - ❖ The resistor with the lower resistance will have higher currents.
 - ❖ Even very high resistances will have some currents.

RESISTORS IN PARALLELE CONT...

5. *Household circuits* are wired so that electrical devices are connected in parallel.
6. In a parallel circuit, transfer of **power** will be the **greatest** with the **resistor with the smallest resistance** according to $P = V^2 / R$
7. In a parallel circuit, the **greatest current** will flow through the **resistor with the smallest resistance** and according to $P = VI$ will also have the **greatest transfer of power**.

Resistors in parallel

When current passes through parallel resistors, it splits up according to the ratios of the resistors.

$$i_b = \frac{I \times R_{\parallel}}{R_b}$$

Where:

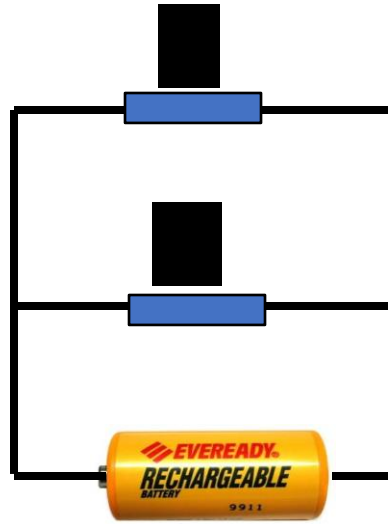
i_b = current in branch

I = total current

R_b = resistor in branch

$R_{\parallel}$ = parallel resistance

Resistors in parallel (special case): for only two resistors



$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$



$$R_p = \frac{R_1 \times R_2}{R_1 + R_2}$$

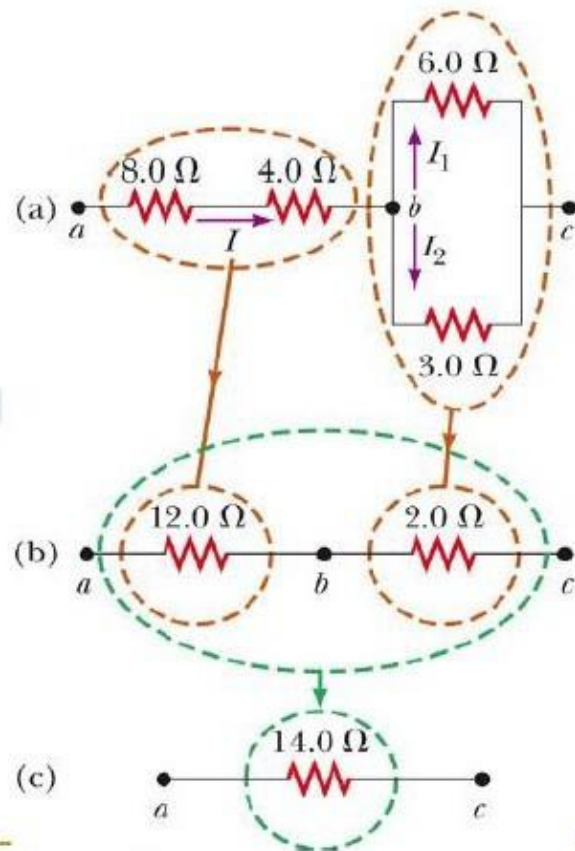
Product

Sum

PART 4: SERIES AND PARALLEL NETWORKS

Equivalent Resistance, Summary

Consider the figure below:



- ❖ The $8.0\ \Omega$ and $4.0\ \Omega$ resistors are in series and can be replaced with their equivalent, $12.0\ \Omega$.
- ❖ The $6.0\ \Omega$ and $3.0\ \Omega$ resistors are in parallel and can be replaced with their equivalent, $2.0\ \Omega$.
- ❖ These equivalent resistances are in series and can be replaced with their equivalent resistance, $14.0\ \Omega$.

Hints on solving Electric Circuits with Combination Circuits

- ❖ When analyzing combination circuits, it is critically important to have a solid understanding of the concepts that pertain to both series circuits and parallel circuits.
- ❖ Since both types of connections are used in combination circuits, the concepts associated with both types of circuits apply to the respective parts of the circuit.

The Concepts of EMF, Pd & Voltage

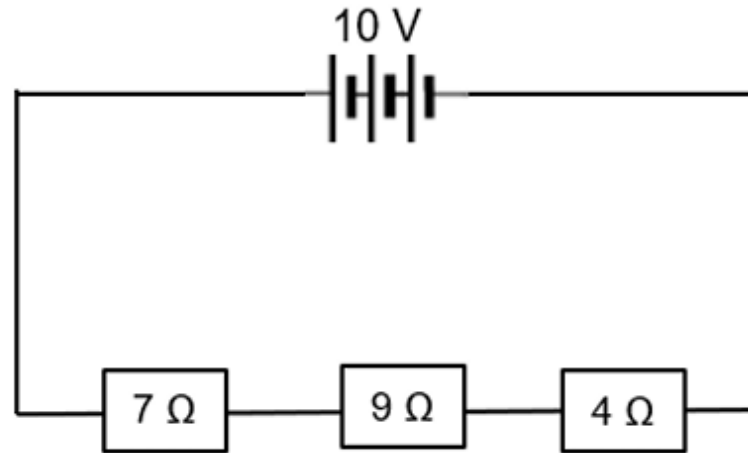
- There is a need to impress on learners that *emf* is a potential difference which is produced by the conversion of other forms of energy (chemical, kinetic, ...) by an *electro pump*, a device which uses non-electrical energy to maintain a potential difference between its terminals.
- The *emf* is produced whether or not the pump is delivering current.
- Furthermore, when delivering *current* I , the electro pump will, by virtue of its *internal resistance* r , have a *pd* V across its terminals *which is less than the emf*, such that

$$Pd (V) = \varepsilon - Ir$$

With Ir representing the *pd inside* the source of *emf*. *This particular relationship is understood by very few learners.*

ACTIVITY 1 :GR 10-11 ASPECTS

1. Consider the following circuit diagram.



1.1 Explain why this is a series circuit? (2)

1.2 Calculate the:

1.2.1 total resistance of a circuit. (3)

1.2.2 total current in a circuit. (3)

1.2.3 potential difference across the...

i. 7 Ω resistor. (3)

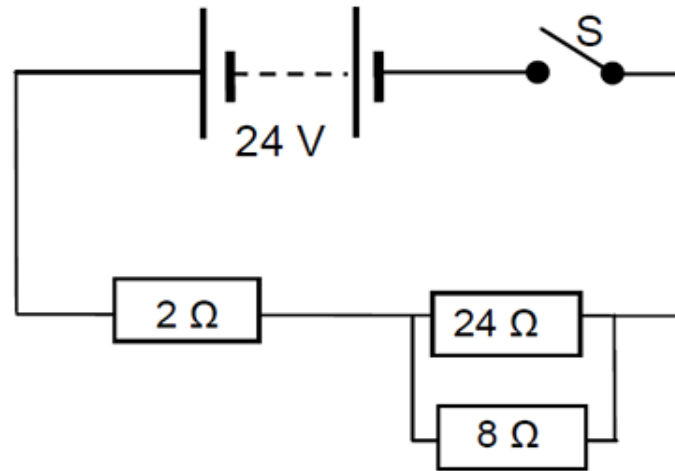
ii. 9 Ω resistor. (2)

iii. 4 Ω resistor. (2)

[15]

ACTIVITY 1 :GR 10-11 ASPECTS

1. Consider the following circuit diagram.



- 1.1 Define the term current in words. (2)
- 1.2 Calculate the equivalent resistance of the resistors connected in parallel. (2)
- 1.3 Calculate the total resistance of the circuit. (2)

Switch **S** is now closed.

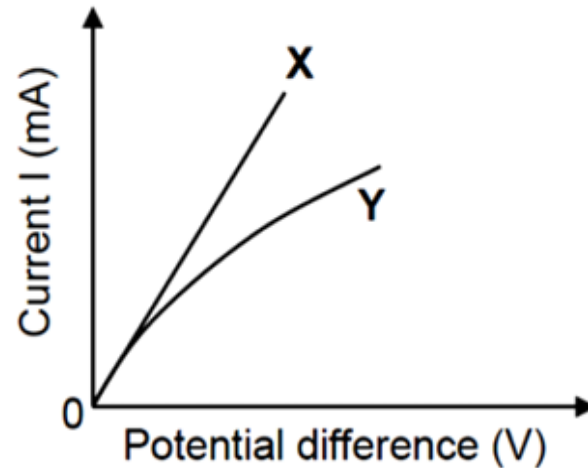
Calculate the:

- 1.4 total current of the circuit. (3)
- 1.5 potential difference across the 2 Ω resistor. (3)
- 1.6 current across the 8 Ω resistor. (3)

[15]

ACTIVITY 1 :GR 10-11 ASPECTS

1. The two graphs alongside show the relationship between current and potential difference for two different conductors, X and Y.



- 1.1 State Ohm's law in words. (2)
- 1.2 Which ONE of the two conductors, X or Y, is ohmic? Refer to the graph and give a reason for the answer. (2)
2. A kettle is marked 240 V; 1500 W.
- 2.1 If the kettle takes 3 minutes to boil some water, calculate the amount of electrical energy transferred to the kettle. (3)
- 2.2 Calculate the resistance of the kettle when operating according to the above specifications. (3)

ACTIVITY 1 :GR 10-11 ASPECTS

A 220 V washing machine draws a current of 20 A for 2 minutes. The price of electricity is R1,15 per kWh.

- 5.1 Define *power* in words. (2)
- 5.2 Calculate the cost to operate the washing machine for two minutes. (6)
- [8]**



(Electric Circuits)

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LESSON 2

GR 12 CONCEPTS

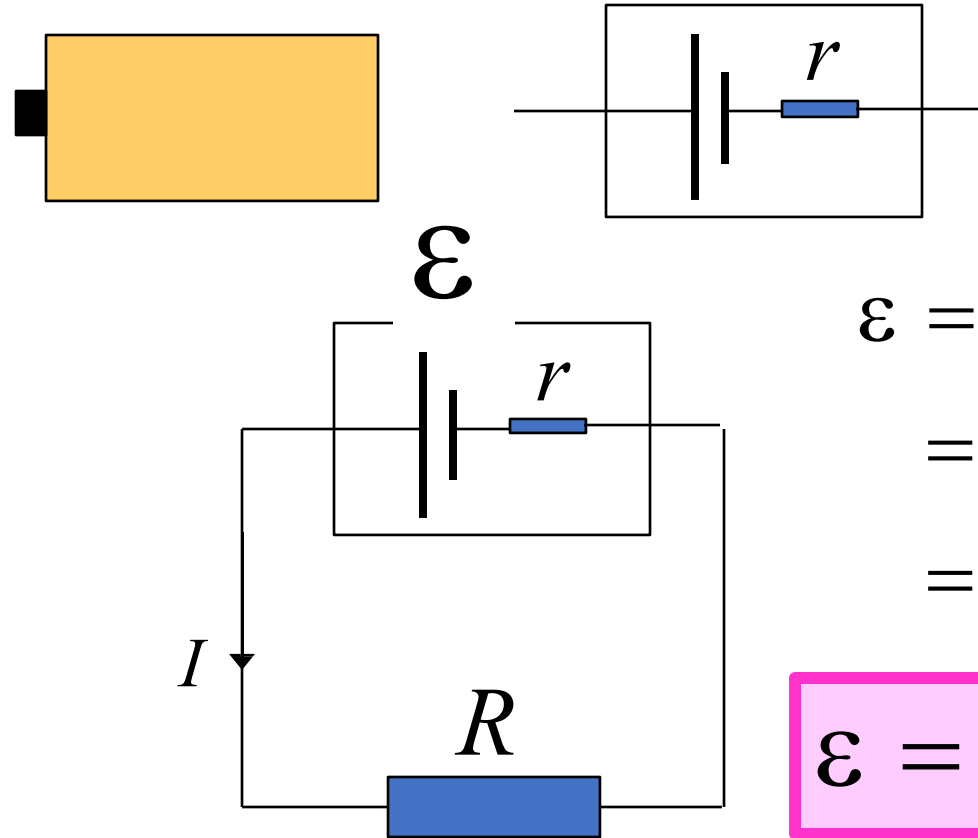
Expose learners to the formulae and their interpretations

ELECTRIC CIRCUITS/ELEKTRIESE STROOMBANE

$R = \frac{V}{I}$	$\text{emf } (\varepsilon) = I(R + r)$ $\text{emk } (\varepsilon) = I(R + r)$
$R_s = R_1 + R_2 + \dots$ $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$	$q = I\Delta t$
$W = Vq$ $W = VI\Delta t$ $W = I^2 R \Delta t$ $W = \frac{V^2 \Delta t}{R}$	$P = \frac{W}{\Delta t}$ $P = VI$ $P = I^2 R$ $P = \frac{V^2}{R}$

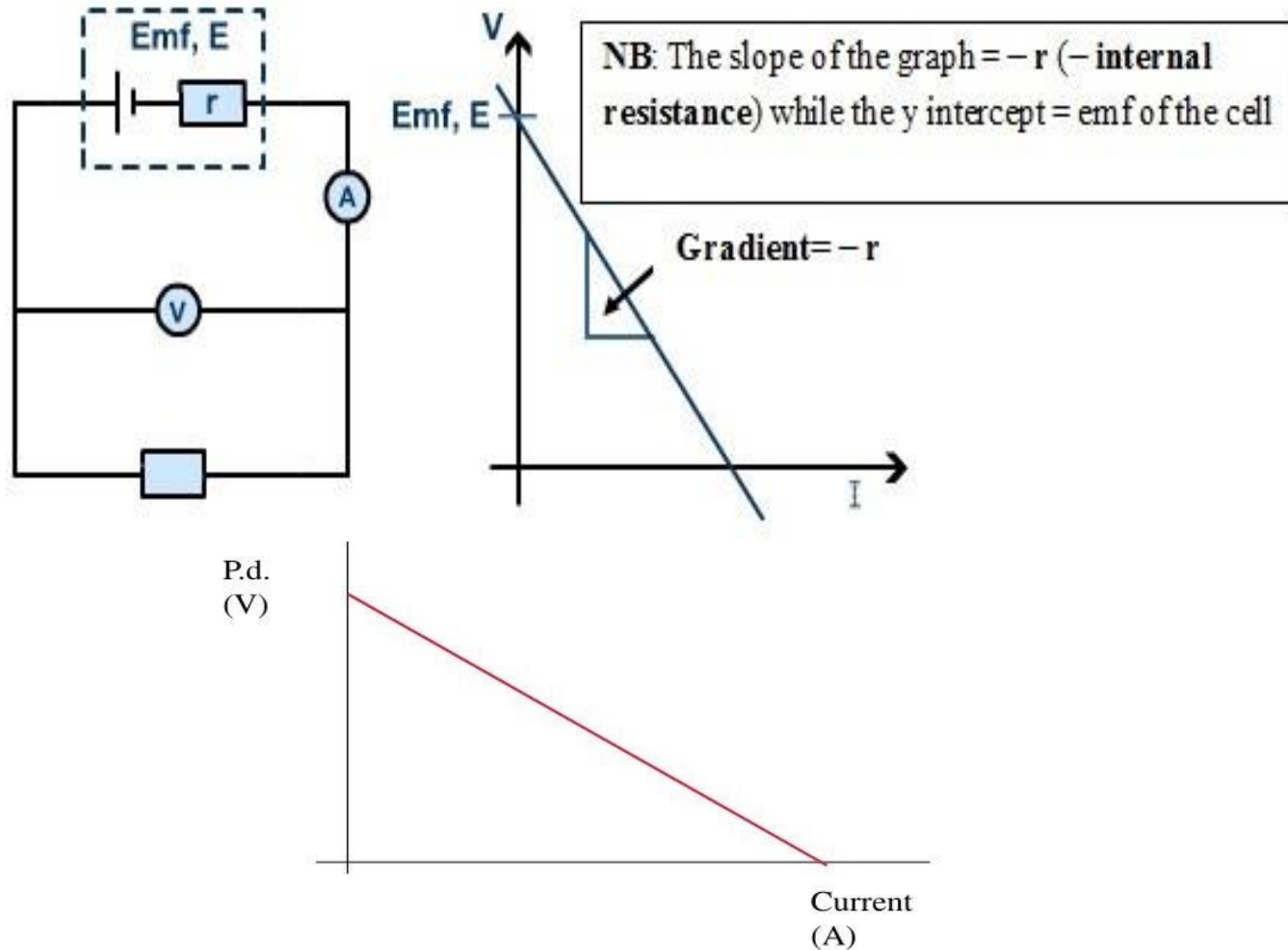
Internal Resistance (Grade 12)

A real battery has internal resistance.



$$\begin{aligned}\varepsilon &= pd + Ir \\ &= IR + Ir \\ &= I(R + r)\end{aligned}$$

$$\varepsilon = I(R + r)$$



$$\mathcal{E} = V + Ir$$

$$\mathcal{E} - Ir = V$$

$$V = -Ir + \mathcal{E}$$

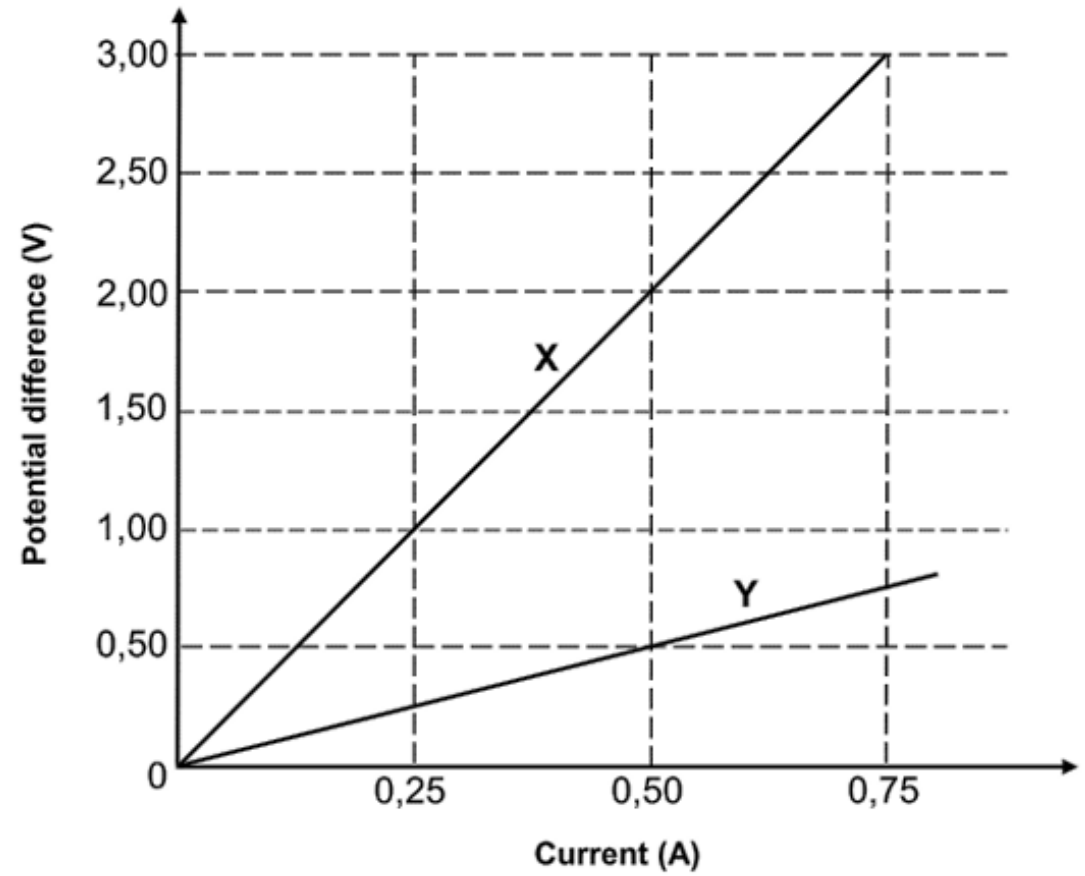
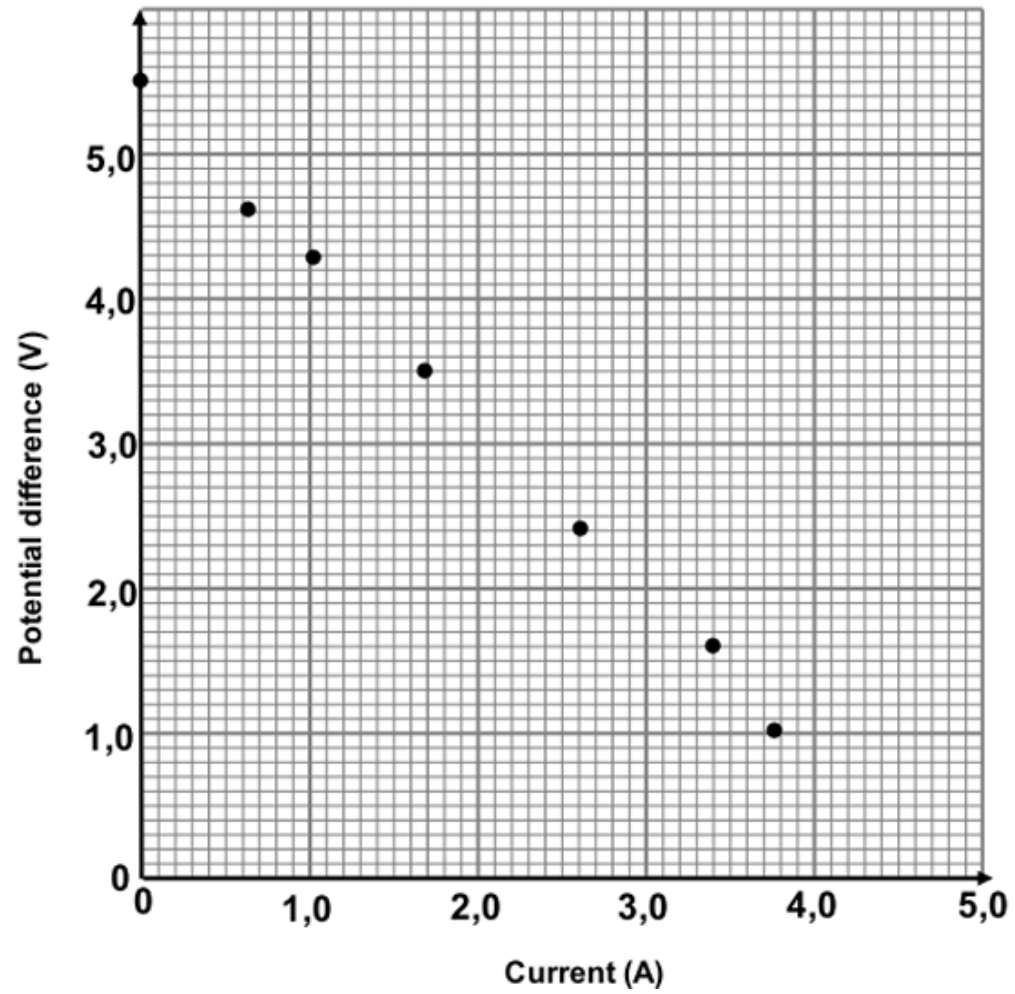
$$V = -rI + \mathcal{E}$$

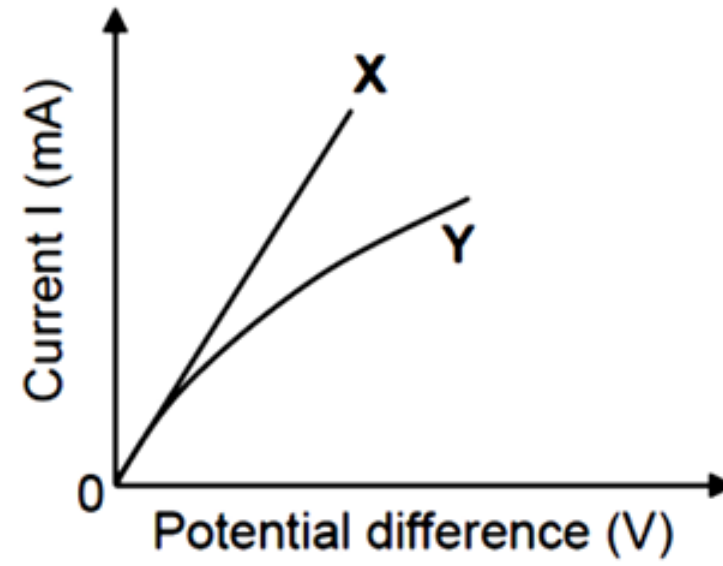
$$y = mx + c$$

The graph is a straight line, of the form $y = mx + c$. We can make the equation for internal resistance $V = -rI + E$. There are three features on the graph that are useful:

- The intercept on the y-axis tells as the emf.
- The intercept on the x-axis tells us the maximum current the cell can deliver when the p.d. is zero, i.e. a dead short circuit.
- The negative gradient tells us the internal resistance.

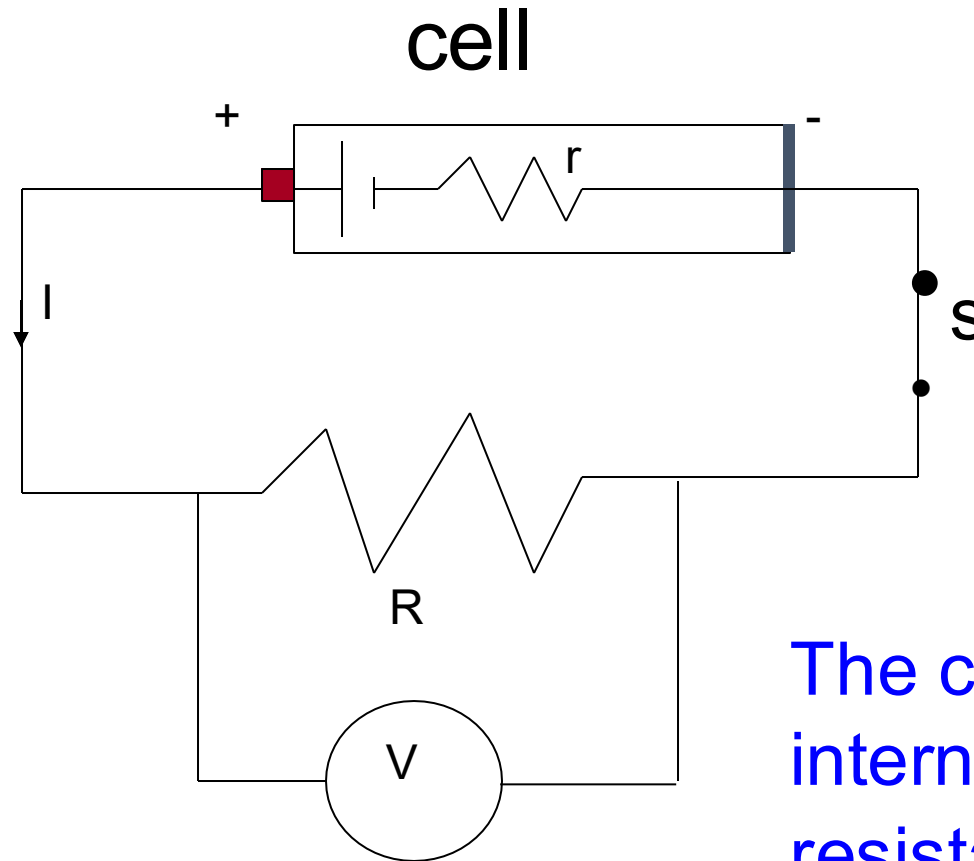
Graph of potential difference versus current





Electric Circuits - Core Concepts

- The **emf** of a cell is a measure of the amount of energy transferred /work done *per coulomb of charge* passing through the battery.
- When a circuit is **closed** and a current flows (pass through), there is also **resistance in the cell**.
- Some of the energy supplied by the cell must therefore be used to move the charge through the cell.

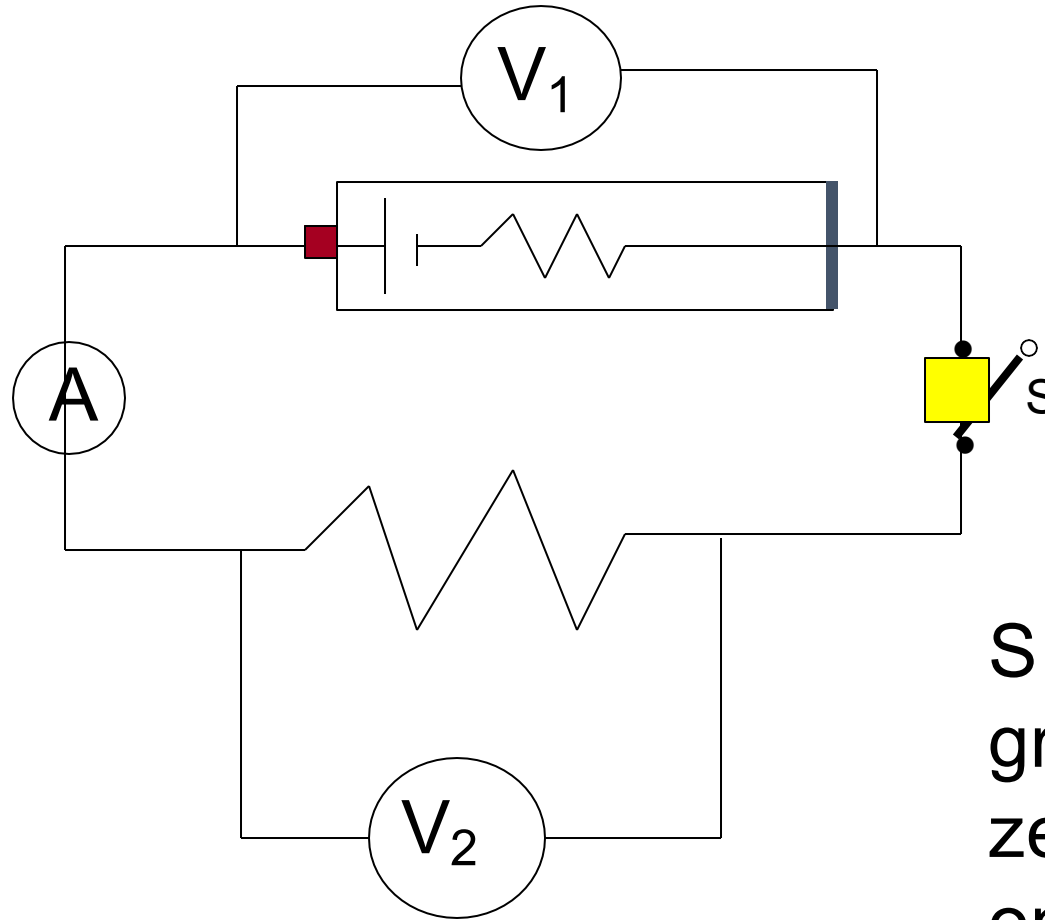


The cell has an internal resistance $r \Omega$.

Electric Circuits - Core Concepts Cont...

- To determine the *internal resistance* r , of the cell it is therefore necessary to make **two measurements**.
- One with the **switch open** and one with the **switch closed**.

With S open – reading on A and V_2 is Zero but there is a greater than zero reading – emf of cell on V_1 .



S closed – greater than zero readings on A, V_1 & V_2 .

Electric Circuits- Core Concepts Cont...

S open:

$$V_1 = \text{emf} = 1,5 \text{ V}$$

$$V_2 = 0 \text{ V}$$

$$A = 0 \text{ A}$$

'Lost volts' is the energy required to move the current through the cell.

S closed:

$$V_1 = \text{ext. p.d.} = 1,2 \text{ V}$$

$$V_2 = 1,2 \text{ V}$$

$$A = 1,2 \text{ A}$$

$$R = 1 \Omega$$

$$V_{\text{emf}} - V_1 = \text{'lost volts'}$$

$$1,5 \text{ V} - 1,2 \text{ V} = 0,3 \text{ V}$$

Electric Circuits- Core Concepts Cont...

- With the switch open, there will be A ZERO reading on the voltmeter V_2 .
- A voltmeter across the cell, when the switch is open will read the emf = V_1 .
- With the switch closed, the voltmeter will read the difference between the **emf** and the energy lost in the cell (due to internal resistance).

Useful formulae

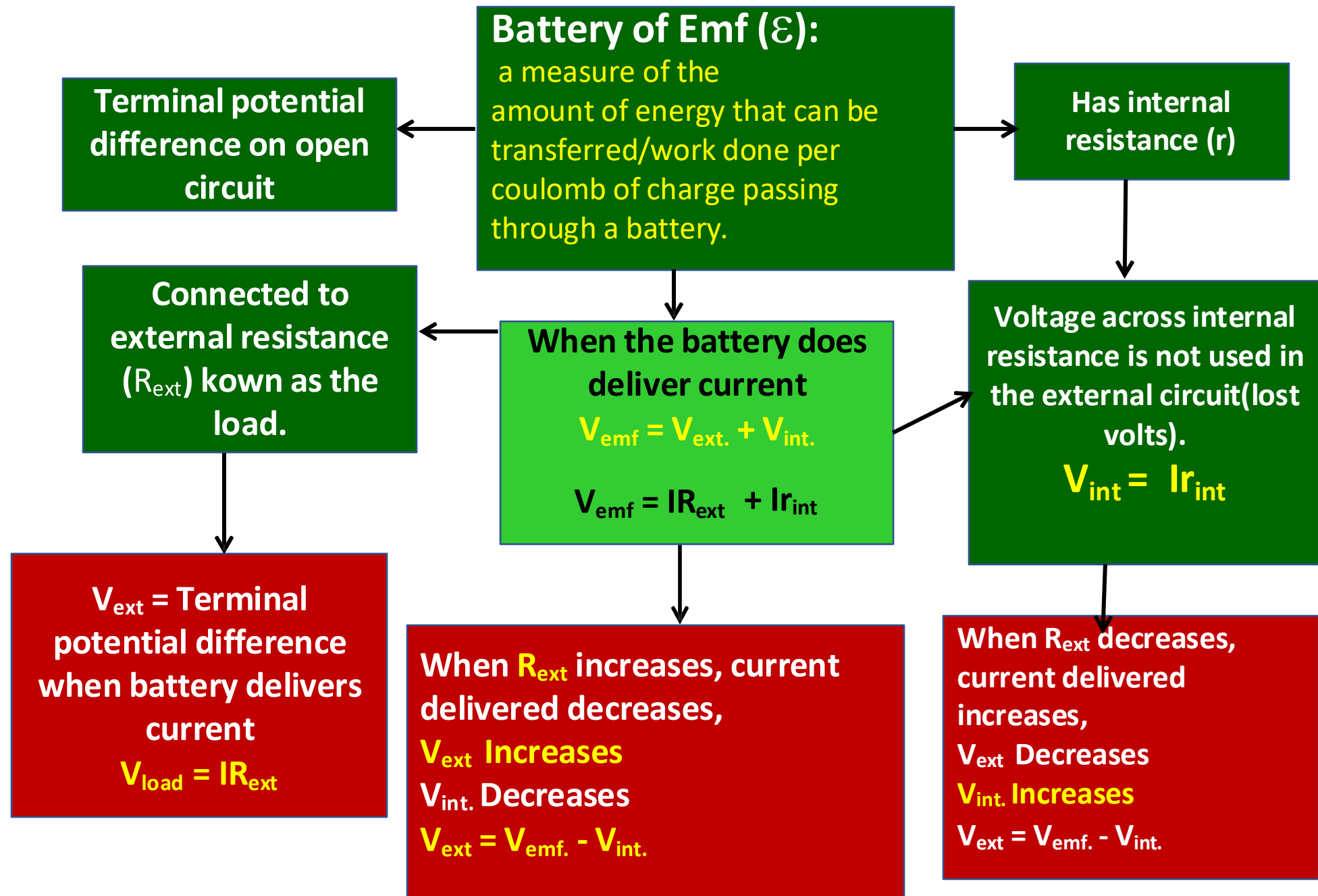
There are 3 important formula to calculate current – be careful how you use them!

$$I_{\text{tot}} = \frac{V_{\text{emf}}}{R_{\text{tot}}} \quad I_{\text{tot}} = \frac{V_{\text{ext.}}}{R_{\text{ext.}}} \quad I_{\text{tot}} = \frac{V_{\text{int.}}}{r_{\text{int.}}}$$

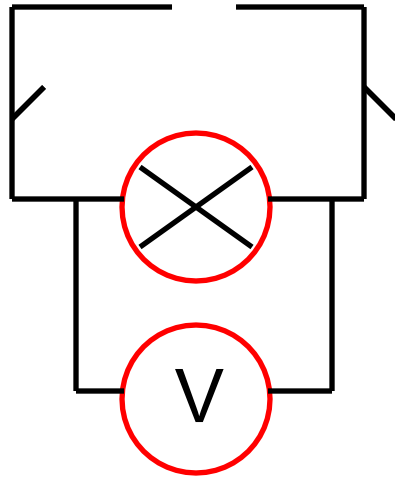
$$V_{\text{emf}} = V_{\text{ext.}} + V_{\text{int.}} \quad \&$$

$$R_{\text{tot}} = R_{\text{ext.}} + r_{\text{int.}} \quad \&$$

$$IR_{\text{tot}} = IR_{\text{ext}} + Ir_{\text{int}}$$



Electrical energy equations



Energy is transferred to the lamp & equals work done in moving charge against the resistance of the lamp.

The potential difference is V , the current is I & Q charge passes through.

Using the equations $W = VQ$, $Q = It$ & $V = IR$

We can deduce the energy equations:

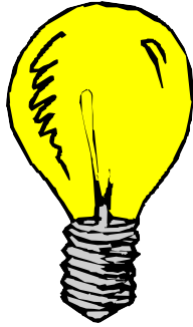
$$W = VIt$$

$$W = I^2Rt$$

$$W = \frac{V^2t}{R}$$

These equations can be used to determine the electrical energy transferred to an electrical equipment in a given time interval.

Electrical power



Written on the bulb is: 220 V & 100 W

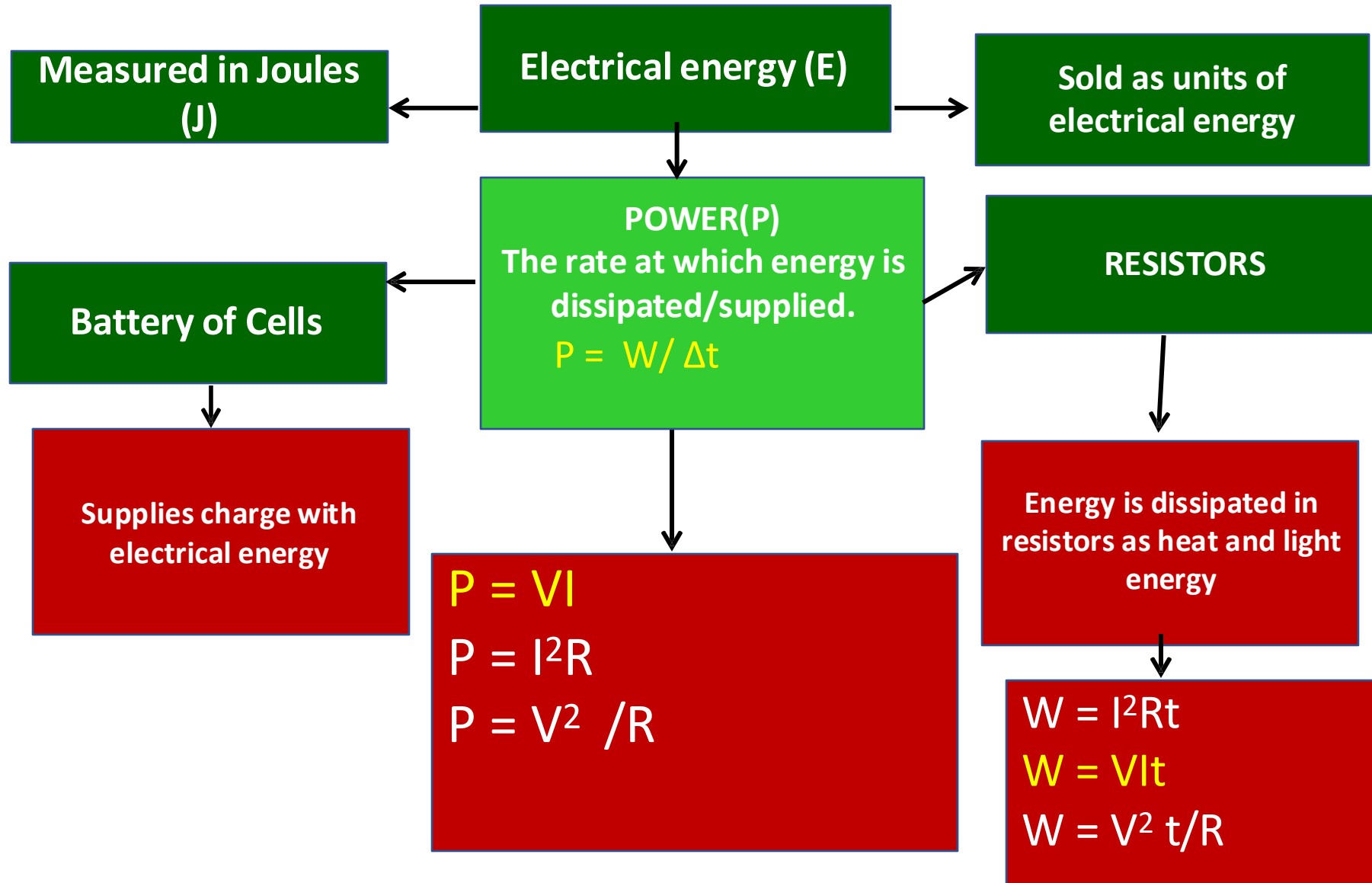
The 100 W indicates the electrical power of the globe.

Power is the rate at which electrical work is done, or the rate at which energy is transferred to the bulb.

Using the energy equations we can deduce the power equations:

$$P = \frac{W}{\Delta t} \quad P = VI \quad P = I^2 R t \quad P = \frac{V^2}{R}$$

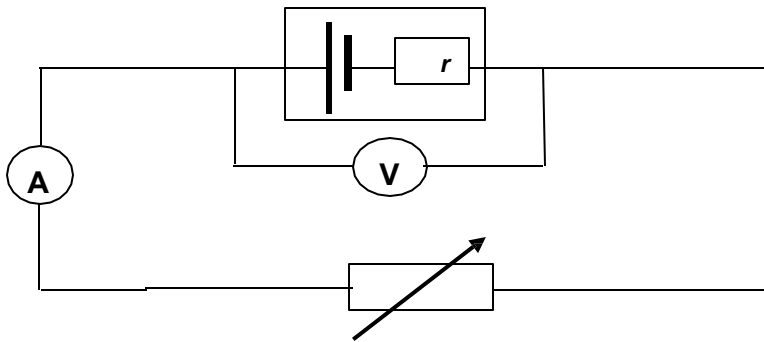
Most pieces of electrical equipment have the potential difference and the power required written on the back of the apparatus.



ACTIVITY 2 : CIRCUITS WITH INTERNAL RESISTANCE

1. Learners conduct an investigation to determine the emf and internal resistance (r) of a battery.

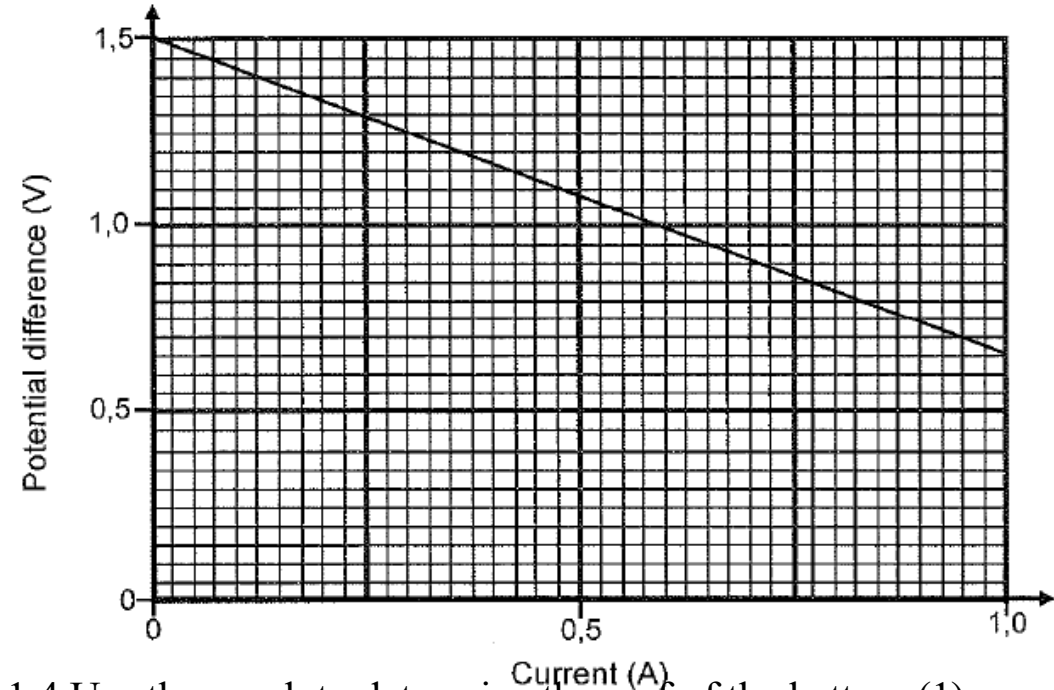
They set up a circuit as shown in the diagram below and measure the potential difference using the voltmeter for different currents in the circuit.



The results obtained are shown in the graph .

1. Which ONE of the measured quantities is the dependent variable? (1)
2. Which ONE of the measured quantities is the independent variable? (1)
3. Write down ONE variable that must be kept constant during this investigation. (1)

Graph of potential difference versus current



1.4 Use the graph to determine the emf of the battery. (1)

4.4 Calculate the gradient of the graph. (3)

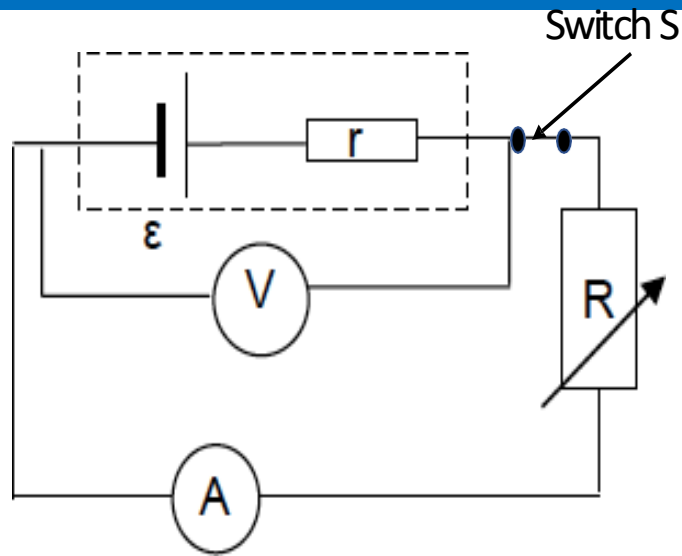
4.5 Which physical quantity is represented by the magnitude of the gradient of the graph? (1)

4.6 How does the voltmeter reading change as the ammeter reading increases? Write down **INCREASES**,

DECREASES or **REMAINS THE SAME**. Use the formula

emf = IR + Ir to explain the answer. (3)

ACTIVITY 2 : CIRCUITS WITH INTERNAL RESISTANCE



Suppose the voltmeter reading is 1,5 V when the switch S is open and 1,2 V when the switch is closed. The ammeter reading is 0,30A.

1. Define the terms :

1. Emf.

(2)

2. Internal resistance.

(2)

1.4 Terminal Potential difference.

(2)

2. Determine the value of

1. Emf.

(1)

2. Terminal potential difference.

(1)

3. Internal resistance.

(3)

2.4 The resistance of R.

(3)

2.4 The Resistance of R is now increased. State what will happen to the voltmeter reading? Write down only INCREASES, DECREASES OR REMAIN THE SAME. Explain your answer fully.

(4)

[18]

Solving Electric Circuits Problems

- Learners will be expected to Solve problems involving current, voltage and resistance for circuits containing arrangements of resistors in series and in parallel.
- Solve/calculate problems involving:
 - ◆ Circuits, including the concept of power
 - ◆ The concept of electrical energy
- The cost of electricity usage, given the power specifications of appliances used and the duration if the cost of 1 kWh is given

SUMMARY: SERIES AND PARALLEL NETWORKS

1. Ohm's Law governs the relationship between current and potential difference for a circuit element at constant temperature. Mathematically we write $R = \frac{V}{I}$
2. Conductors that obey Ohm's Law are called ohmic conductors; those that do not are called non-ohmic conductors.
3. Ohm's Law can be applied to a single circuit element or the circuit as a whole (if the components are ohmic).
4. The equivalent resistance of resistors in series (R_s) can be calculated as follows:

$$R_s = R_1 + R_2 + R_3 + \dots + R_n$$

5. The equivalent resistance of resistors in parallel (R_p) can be calculated as follows:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

6. Real batteries have an internal resistance.
7. The potential difference (V) of the battery is related to its emf (ϵ) and internal resistance (r) by:
 $\epsilon = V_{\text{load}} + V_{\text{internal resistance}}$ or $\epsilon = IR_{\text{ext}} + Ir$.
8. The external resistance in the circuit is referred to as the load.
9. Define power as the rate at which work is done. Mathematically we write $P = W/\Delta t$.
10. When solving circuit problems involving the concepts of power and electrical energy, we use the following:
 $P = VI$, $P = I^2R$ or $P = V^2/R$

NOTE THE FOLLOWING WITH REFERENCE TO CALCULATIONS

- ❖ Always use **different subscripts** to identify the various components in calculations.
- ❖ The current is the same at all points around the circuit, except where the current splits in the parallel circuit.
- ❖ In **parallel circuits**, where the current divides, the voltage is the same at all points.
- ❖ In **series circuits**, the current is the same everywhere, but the voltage divides.
- ❖ The question: “*Calculate the current passing through the circuit*” may be asked in a variety of ways – and they all mean the same thing:
 - Calculate the current through the ammeter.
 - Calculate the current through the battery.
 - Calculate the current through the light bulb/resistor.
 - What is the ammeter reading?

APPROACH: ELECTRIC CIRCUITS

The following steps may be used when solving circuit calculations:

1. If a schematic diagram is not provided, sketch the circuit and fill in the data.
2. Identify whether there is internal resistance or not?
3. If there is No internal resistance, the terminal potential difference on open circuit is equal to the emf of the battery or cell

4. If there is an internal resistance,

$$V_{\text{emf}} = V_{\text{ext.}} + V_{\text{int.}}$$

$$V_{\text{emf}} = IR_{\text{ext}} + Ir_{\text{int}}$$

(When the battery delivers current).

5. If there is an internal resistance, there are TWO circuits: **internal** and **external** circuits, summed up the formula

$$\mathcal{E} = I (R + r)$$

- Calculate the emf (E) of the battery – if not given.
- Calculate the internal resistance (r) of the battery - if not given.
- Calculate the Total Current - if not given.

6. For the external circuit, ask yourself how many resistors constitute this circuit and how are they connected. (In most exam questions, the external circuit is a combination of Series- Parallel circuits).

- If solving Series-Parallel circuits, solve the Parallel parts first, by Calculating the total resistance (R) of the resistors in parallel – i.e. the equivalent resistance of the grid
- Calculate the total external resistance (**R_{ext}**) of the circuit
- Calculate the total resistance of the circuit ($R_{ext} + r_i$).

7. Apply the formula (**$R + r = \frac{E}{I}$**) **OR** **$E = I(R + r)$**

8. Calculate the current through the circuit.

9. Calculate the answers to questions relating to different components of the circuit.

- Make sure you know which **I** , **V** and **R** applies to that component of the circuit.
- Make use of subscripts to identify the various components.
- Use the Ohm's law equation (**$R = \frac{V}{I}$**) often and appropriately. Most answers will be determined using this equation.
- When using this formula (**$R = \frac{V}{I}$**), it is important to substitute the appropriate values into the equation. For instance, if calculating I_2 , it is important to substitute the **V_2** and the **R_2** values into the equation.

Solving Electric Circuits Problems- Possible Exam Scenarios

A real battery has internal resistance.

$$\varepsilon = I(R + r)$$

$$\varepsilon = pd + Ir$$

$$= IR + Ir$$

$$= I(R + r)$$

When Solving problems involving current, voltage and resistance for circuits containing arrangements of resistors in series and in parallel, you will usually be given one of the following cases below:

- ❑ **CASE 1:** *Emf* (E), Internal resistance (r) of the battery and total current in the circuit are KNOWN.
- ❑ **CASE 2:** *Emf* (E), Internal resistance (r) of the battery are KNOWN but total current in the circuit is UNKNOWN.
- ❑ **CASE 3:** *Emf* (E) of the battery is KNOWN, Internal resistance (r) of the battery is UNKNOWN but total current in the circuit is KNOWN. (Sometimes in terms of Power rating OR Potential Difference across of one of the resistors in the external circuit using $P = VI$ **OR** $R = V/I$ **OR** using $I_{\text{tot}} = V_{\text{tot}}(\text{Emf})/R_{\text{tot}}$)
- ❑ **CASE 4:** *Emf* (E) of the battery is UNKNOWN, Internal resistance (r) of the battery is KNOWN and total current in the circuit is also KNOWN (Sometimes in terms of Power rating **OR** Potential Difference across of one of the resistors in the external circuit using $P = VI$ **OR** $R = V/I$ **OR** using $I_{\text{tot}} = V_{\text{int}}(\text{Lost volts})/r_i$)
- ❑ **CASE 5:** *Emf* (E), Internal resistance (r) of the battery and total current in the circuit are UNKNOWN.
- ❑ **CASE 6:** *Emf* (E) IS KNOWN, But Internal resistance (r) of the battery and total current in the circuit are UNKNOWN.

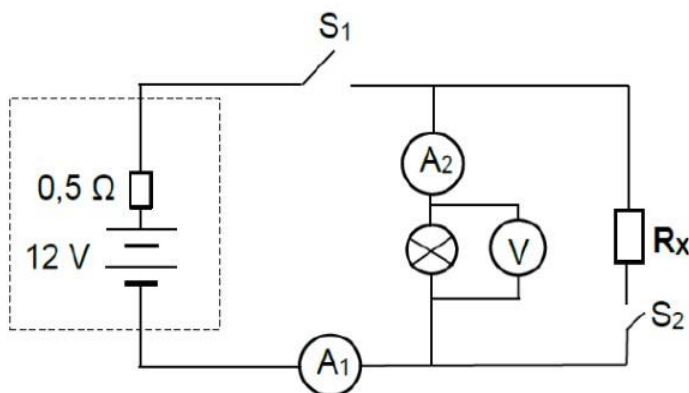
Solving Electric Circuits Problems – Key Words and Implications

- When dealing with electric circuits with *internal resistance* r , of the cell it is therefore necessary to make **two measurements** (i.e. One with the **switch open** and one with the **switch closed**.
- With the MAIN SWITCH open, there will be A ZERO reading on the voltmeter across the resistor of the external circuit.
- A voltmeter across the cell, when the MAIN SWITCH is open will read the *Emf (E) of the battery*.
- With the switch closed, the voltmeter will read the difference between the **emf** and the energy lost in the cell (due to internal resistance) – The Pd.
- **Decreases by/with**, when the MAIN SWITCH is CLOSED the voltmeter will read the *lost volts (due to the internal resistance of the battery)*.
- **Changes to**, when the MAIN SWITCH is CLOSED the voltmeter will read the *potential difference drop (V_{external})*.
- **Changes to**, when the MAIN SWITCH is OPENED the voltmeter will read the *Emf (E) of the battery*.

EXAM TYPE QUESTIONS : ELECTRIC CIRCUITS

QUESTION 8 (Start on a new page.)

- 8.1 A pump is connected to a water tank to pump rainwater into a house. The pump is rated 750W and it is connected to the 240 V main supply.
- 8.1.1 What current will the pump draw when operating on the 240 V supply? (3)
- 8.1.2 The cost of electricity is R1,20 per kilowatt hour. Calculate the cost of using the pump continuously for 20 minutes. (3)
- 8.2 In the circuit represented below, the battery has an emf of 12 V and an internal resistance of $0,5\ \Omega$. The battery is connected as shown to a light bulb and resistor **R_x**, both of unknown resistance. Ammeters have zero resistance and the voltmeter has infinite resistance.



Switch **S₁** and switch **S₂** are initially both open.

- 8.2.1 Define *emf*. (2)
- 8.2.2 What will be the reading on the voltmeter, V, when both switches are open? (2)

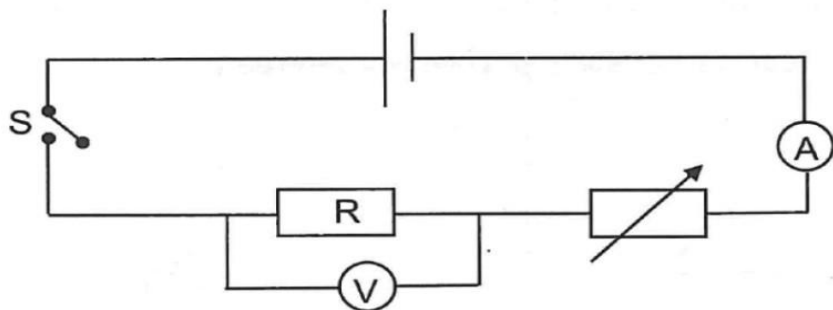
Switch **S₁** is now closed, while **S₂** remains open. Ammeter **A₁** reads 1,6 A.

- 8.2.3 Determine the resistance of the bulb. (4)
- 8.2.4 Calculate the reading on the voltmeter, V. (3)
- 8.2.5 Calculate the rate of energy dissipation in the battery. (3)
- 8.2.6 Switch **S₂** is now closed so that both switches are closed. When switch **S₂** is closed, state whether the following will DECREASE, INCREASE or STAY THE SAME:
- (a) the reading on ammeter **A₁**. Explain your answer. (2)
- (b) the rate of energy dissipation in the battery. (2)
- (c) the reading on voltmeter V. Explain your answer. (2)

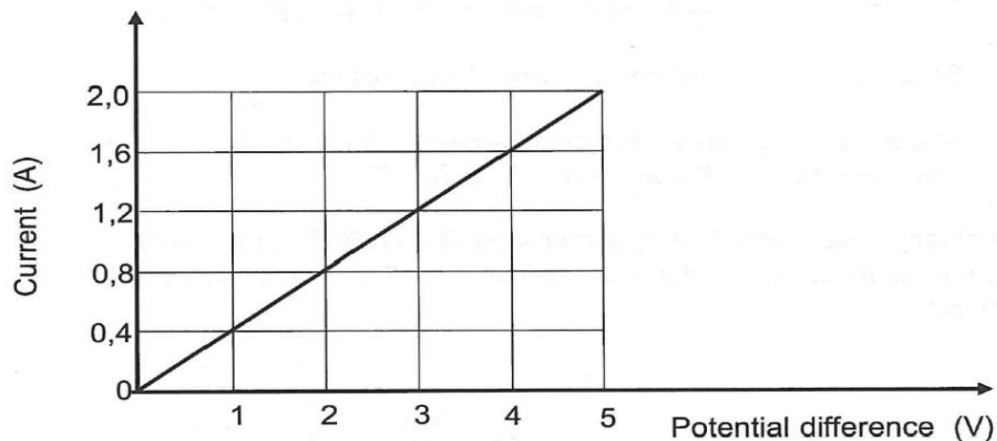
EXAM TYPE QUESTIONS CONTINUED,,

QUESTION 3 (Start on a new page.)

- 3.1 The circuit diagram represented below is used to investigate the relationship between the **current** and the **potential difference** across resistor **R**.



The results obtained are used to draw the graph below:



For this investigation, write down the:

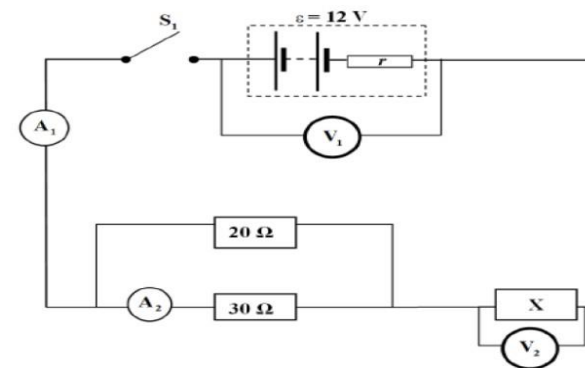
- 3.1.1 Dependent variable (1)
- 3.1.2 Control variable (1)
- 3.1.3 Conclusion that can be drawn from the graph (2)

The resistance of resistor **R** must be determined *using the gradient of the graph*.

- 3.1.4 Calculate the resistance of resistor **R**. (3)

- 3.2 An electric circuit is set up as shown in the diagram below. The resistances of the switch, ammeters and connecting wires are negligible. The voltmeters have very high resistance.

The battery has an emf (ϵ) of 12 V and has significant internal resistance (r).



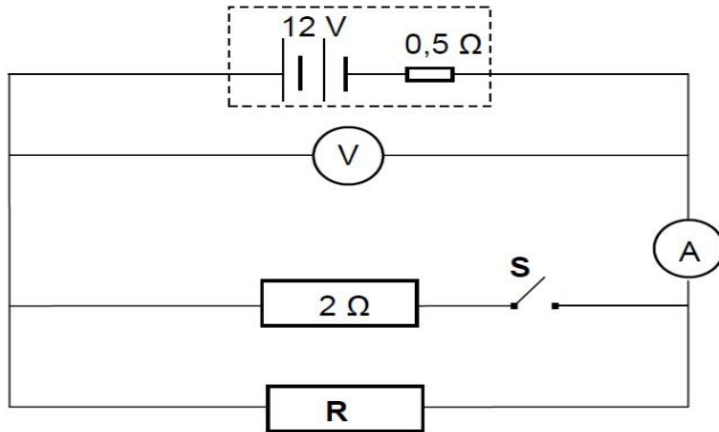
The switch 1 **S** is **CLOSED**. The ammeter **A₂** reads **0,2 A** and the voltmeter **V₂** reads **5,5 V**.

- 3.2.1 Define *emf*. (2)
- 3.2.2 Calculate the reading on ammeter **A₁**. (3)
- 3.2.3 Calculate the resistance of resistor **X**. (3)
- 3.2.4 Calculate the total external resistance of the circuit. (3)
- 3.2.5 Calculate the internal resistance (r) of the battery. (3)
- 3.3 Resistor **X** is replaced by a new resistor of greater resistance than that of **X**.
 - 3.3.1 Will the reading on the voltmeter **V₁** connected across the terminals of the battery **INCREASE**, **DECREASE** or **REMAIN THE SAME**? (1)
 - 3.3.2 Explain your answer to Question 3.2.1, making reference to **relevant formulae**. (3)

EXAM TYPE QUESTIONS(NSC NOV 2018)

QUESTION 8 (Start on a new page.)

The battery in the circuit diagram below has an emf of 12 V and an internal resistance of 0,5 Ω . Resistor **R** has an unknown resistance.



8.1 What is the meaning of the following statement?

The emf of the battery is 12 V.

The reading on the ammeter is 2 A when switch **S** is OPEN.

8.2 Calculate the:

8.2.1 Reading on the voltmeter

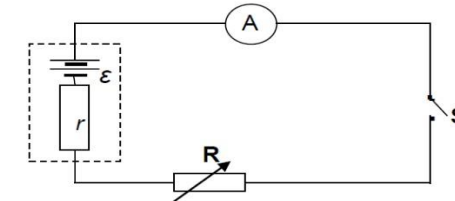
8.2.2 Resistance of resistor **R**

Switch **S** is now CLOSED.

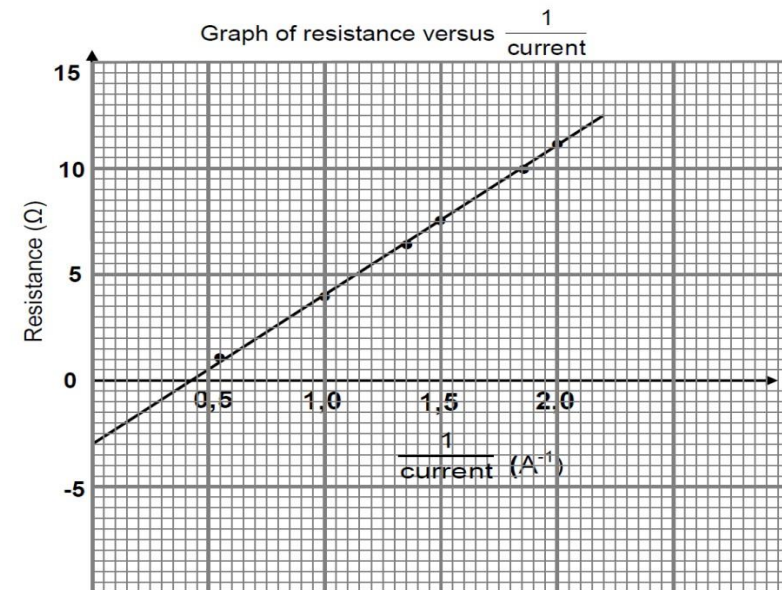
8.3 How does this change affect the reading on the voltmeter? Choose from: INCREASES, DECREASES or REMAINS THE SAME. Explain the answer.

QUESTION 9 (Start on a new page.)

Learners perform an experiment to determine the emf (ϵ) and the internal resistance (r) of a battery using the circuit below.



The learners use their recorded readings of current and resistance, together with the equation $R = \frac{\epsilon}{I} - r$, to obtain the graph below.



(2)

(3)

(2)

9.1 Which variable has to be kept constant in the experiment?

(1)

9.2 Refer to the **graph**.

9.2.1 Write down the value of the internal resistance of the cell.

(2)

9.2.2 Calculate the emf of the battery.

(3)

(4)
[11]

EXAM TYPE QUESTIONS(NSC NOV 2019)

QUESTION 8 (Start on a new page.)

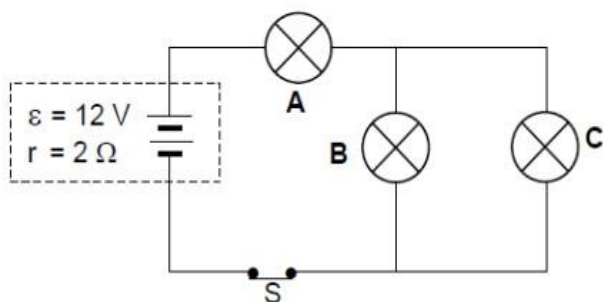
8.1 Three identical light bulbs, **A**, **B** and **C**, are each rated at 6 W, 12 V.

8.1.1 Define the term *power*. (2)

8.1.2 Calculate the resistance of EACH bulb when used as rated. (3)

The light bulbs are connected in a circuit with a battery having an emf (ε) of 12 V and internal resistance (r) of $2\ \Omega$. Refer to the diagram below.

Assume that the resistance of each light bulb is the same as that calculated in QUESTION 8.1.2. Switch S is closed.

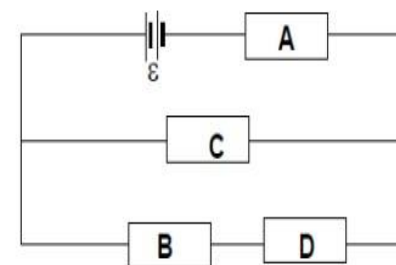


8.1.3 Calculate the total current in the circuit. (5)

8.1.4 Calculate the potential difference across light bulb **C**. (3)

8.1.5 Explain why light bulb **C** in the circuit will NOT burn at its maximum brightness. (3)

8.2 Resistors **A**, **B**, **C** and **D** are connected to a battery having emf (ε) and negligible internal resistance, as shown in the diagram below.



8.2.1 Give a reason why the current in resistor **A** is greater than that in resistor **C**. (2)

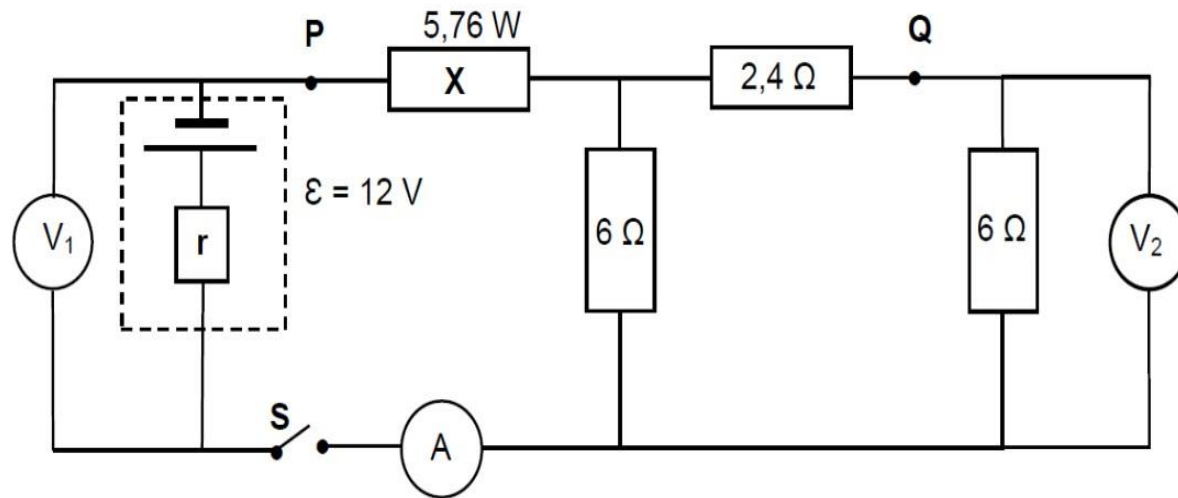
8.2.2 Resistor **C** is removed. How will the current in resistor **B** compare to the current in **A**? Give a reason for the answer. (2)
[20]

NCS 2021 (NOV) : Requirements on Electric circuits-CASE ???

QUESTION 8 (Start on a new page.)

The battery in the circuit shown below has an emf of 12 V and an unknown internal resistance r .

The resistance of the connecting wires and the ammeter is negligible.



Switch **S** is OPEN.

8.1 Write down the reading on:

8.1.1 Voltmeter V_1 (1)

8.1.2 Voltmeter V_2 (1)

Switch **S** is now CLOSED.

The reading on the ammeter is 1,2 A and the power dissipated in resistor **X** is 5,76 W.

8.2 Define the term *power*. (2)

Calculate the:

8.3 Resistance of resistor **X** (3)

8.4 Total EXTERNAL resistance of the circuit (3)

8.5 Reading on voltmeter V_2 (5)

A length of wire of negligible resistance is used to connect point **P** to point **Q** in the circuit.

8.6 How will the reading on voltmeter V_1 be affected?

Choose from INCREASES, DECREASES or NO EFFECT.

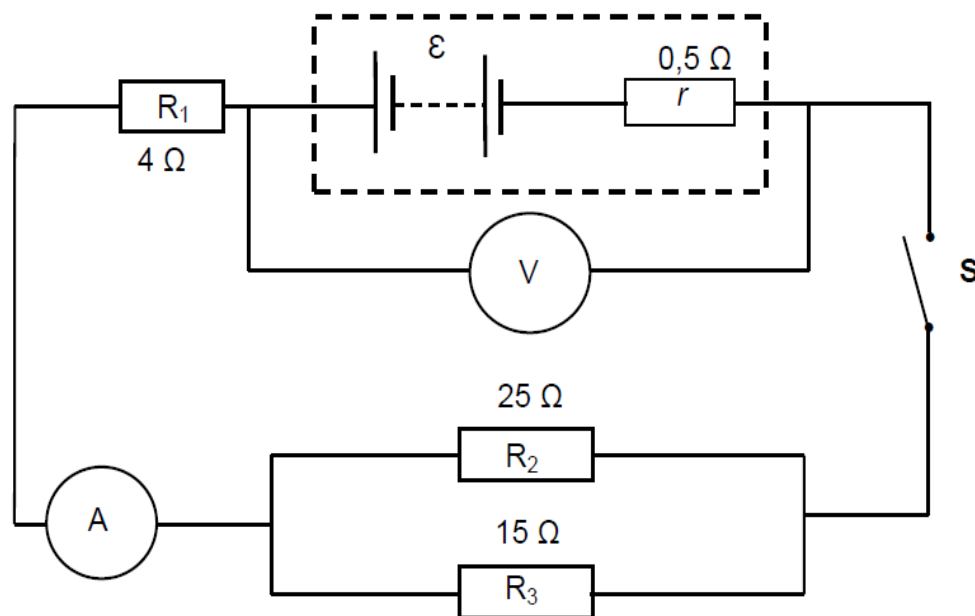
Explain the answer. (4)
[19]

EXAM TYPE QUESTIONS(NSC NOV 2020)

QUESTION 8 (Start on a new page.)

A battery with an internal resistance of $0,5\ \Omega$ and an unknown emf (ϵ) is connected to three resistors, a high resistance voltmeter and an ammeter of negligible resistance, as shown in the circuit diagram below.

The resistance of the connecting wires must be ignored.



8.1 Define the term *emf* of a battery.

(2)

The reading on the voltmeter DECREASES by $1,5\ \text{V}$ when switch **S** is closed.

8.2 Give a reason why the voltmeter reading decreases.

(2)

8.3 Calculate the following when switch **S** is closed:

8.3.1 Reading on the ammeter

(3)

8.3.2 Total external resistance of the circuit

(4)

8.3.3 Emf of the battery

(3)

8.4 A learner makes the following statement:

The current through resistor R_3 is larger than the current through resistor R_2 .

Is this statement CORRECT? Choose from YES or NO. Explain the answer.

(3)

8.5 The $4\ \Omega$ resistor is now removed from the circuit.

How will this affect the emf of the battery? Choose from INCREASES, DECREASES or REMAINS THE SAME.

(1)

[18]

EXAM TYPE QUESTIONS(NSC JUNE 2021)

Physical Sciences/P1

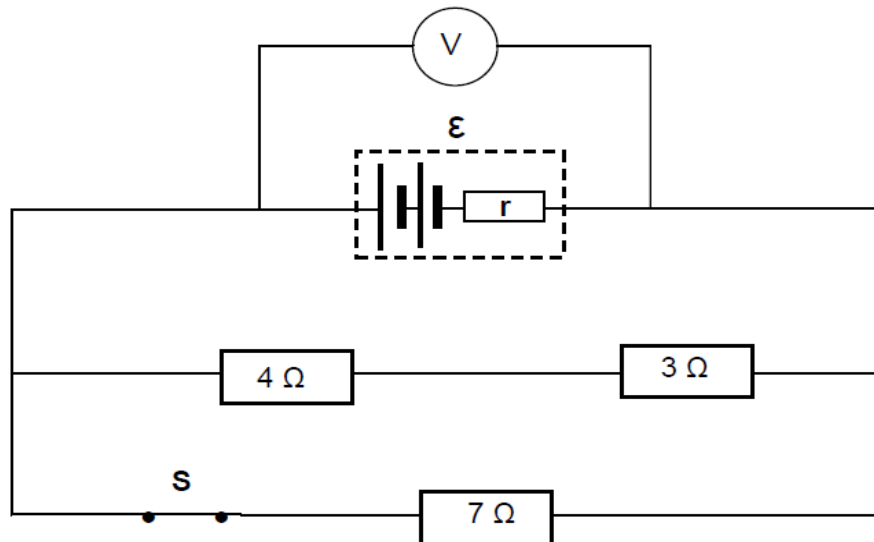
15
SC/NSC

DBE/2021

QUESTION 8 (Start on a new page.)

Three resistors are connected to a battery with an unknown emf and unknown internal resistance r , as shown below.

Ignore the resistance of the connecting wires.



- 8.1 In the definition of the emf of a battery given below, (a) and (b) represent missing words or phrases.

The emf of the battery is the maximum (a) ... supplied by a battery per (b) passing through it.

Write down (a) and (b) in your ANSWER BOOK and next to each the missing word or phrase. (2)

With switch **S** CLOSED, the voltmeter reads 2,63 V.

- 8.2 Calculate the equivalent external resistance of the circuit. (4)

Switch **S** is now OPENED and the voltmeter reads 2,8 V.

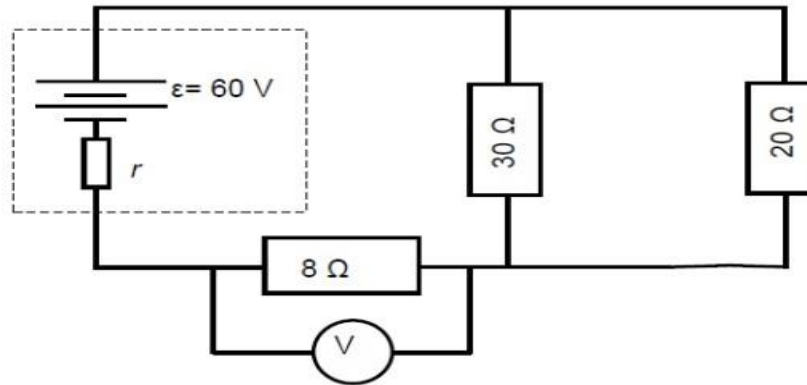
- 8.3 Calculate:

8.3.1 The internal resistance of the battery (8)

8.3.2 The emf of the battery (2)

MARATHON PROBLEM 1

- 9.2 The circuit diagram below shows a battery with an emf (ϵ) of 60 V and an unknown internal resistance r , connected to three resistors.



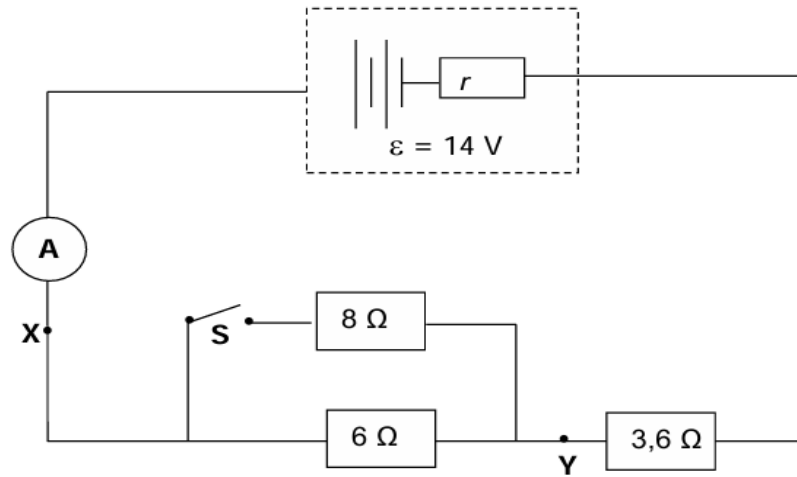
A voltmeter connected across the 8Ω resistor reads 21,84 V.

Calculate the:

- | | | |
|-------|--|-------------|
| 9.2.1 | Current in the 8Ω resistor | (3) |
| 9.2.2 | Equivalent resistance of the resistors in parallel | (2) |
| 9.2.3 | Internal resistance r of the battery | (4) |
| 9.2.4 | Heat dissipated in the external circuit in 0,2 seconds | (3) |
| | | [20] |

MARATHON PROBLEM 2

A battery with an emf of 14 V and unknown internal resistance 'r' is connected in a circuit as shown below.



When the switch **S** is open, the ammeter reading is 1,4 A.

9.1 Calculate the:

9.1.1 Total resistance of the circuit when the switch **S** is open. (3)

9.1.2 Internal resistance of the battery. (3)

9.1.3 Energy dissipated at 6 Ω resistor in three minutes. (3)

9.2 The switch **S** is now closed. Calculate the current through 8 Ω resistor. (6)

9.3 A conducting wire of negligible resistance is now connected between points **X** and **Y**. What effect will it have on the temperature of the battery?

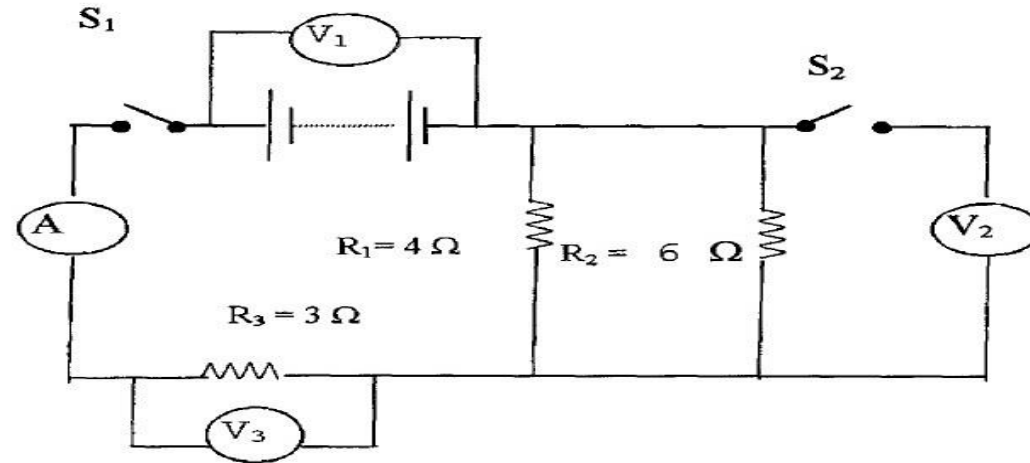
Choose from INCREASE, DECREASE or STAYS THE SAME.

Give a reason for your answer.

(2)
[17]

MARATHON PROBLEM 3

In the following circuit diagram, the battery has an unknown internal resistance r , and its emf is 15 V. Resistors R_1 , R_2 and R_3 are $4\ \Omega$, $6\ \Omega$ and $3\ \Omega$ respectively.



- 3.1 Calculate the resistance of the parallel combination of resistors. (3)
- 3.2 When **only** switch S_1 is **closed**, V_3 reads 7,5 V.
- 3.2.1 Calculate the current in the circuit (reading on ammeter A). (3)
- 3.2.2 Calculate the internal resistance of the battery. (6)
- 3.3 If switch S_2 is now also closed, state whether the following will **INCREASE NOTICEABLY**, **DECREASE NOTICEABLY** or **REMAIN THE SAME**:
- 3.3.1 Voltmeter reading V_2 (2)
- 3.3.2 Ammeter reading A (2)

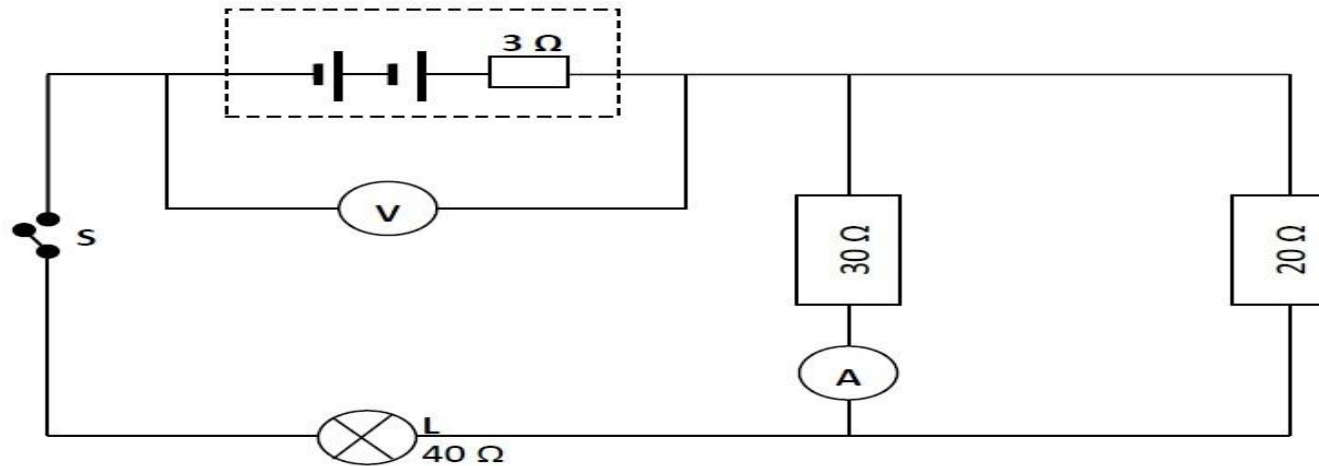
MARATHON PROBLEM 4

INTERNAL RESISTANCE CIRCUIT

QUESTION 1

(GRADE 12)

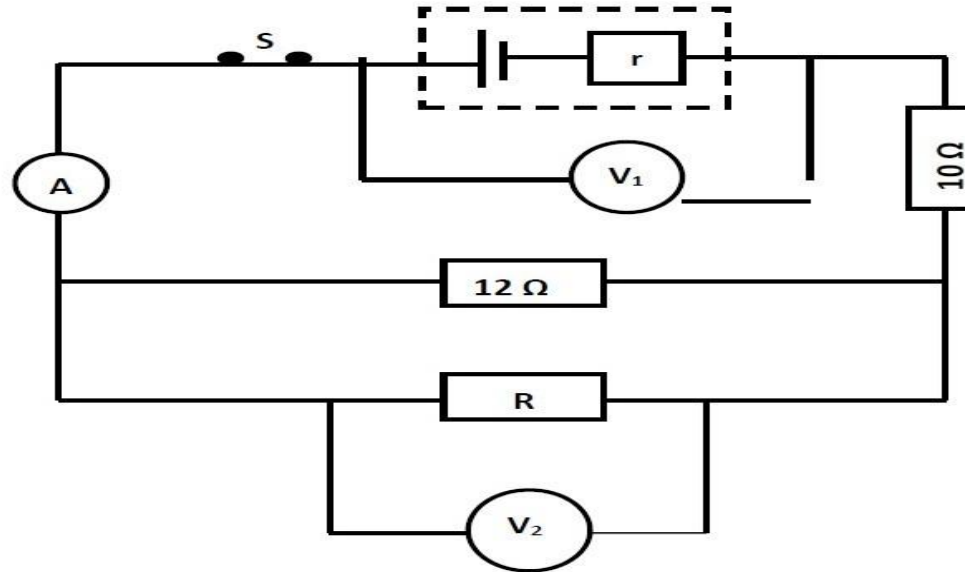
In the circuit diagram below the light bulb L has a resistance of $40\ \Omega$ and the battery has an internal resistance of $3\ \Omega$. The reading on the voltmeter **decreases** with 4, 5 V when the switch S is closed.



- 1.1 Explain why there is a decrease in the voltmeter reading when the switch S is closed. (2)
- 1.2 The switch S is closed. Calculate the:
- 1.2.1 current in the battery (3)
- 1.2.2 emf of the battery (5)
- 1.2.3 reading on the ammeter (A) (3)
- 1.3 The $20\ \Omega$ resistor suddenly burns out. How will the power dissipated in the light bulb change? Write **down INCREASES, DECREASES** or **REMAINS THE SAME** and explain the answer fully. (4)

MARATHON PROBLEM 5

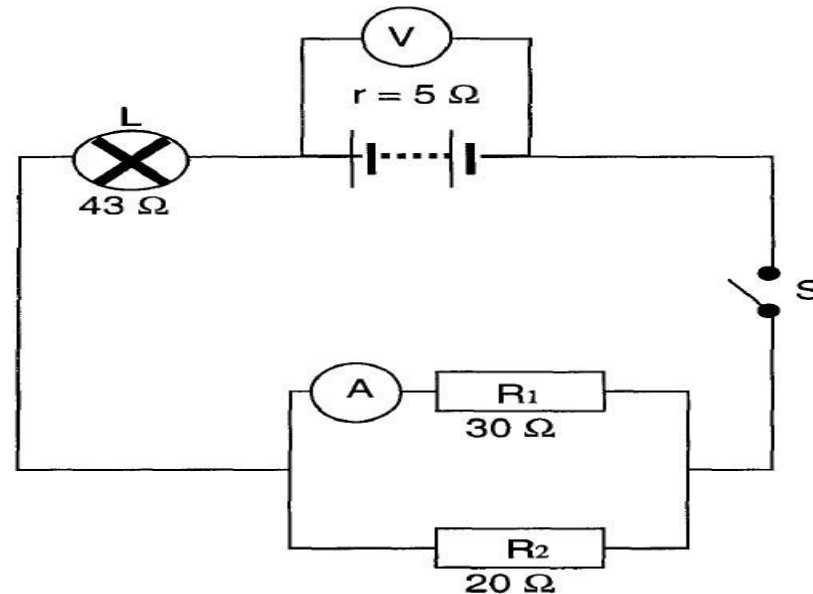
The battery in the circuit, represented in the diagram below, has internal resistance r . When the switch S is closed the reading on voltmeter V_2 is 18 V and resistor R dissipates 13,5 W.



- 4.1 Calculate the resistance of R . (3)
- 4.2 Calculate the reading on the ammeter. (5)
- 4.3 Explain, in words, what is meant by the term *internal resistance*. (2)
- 4.4 Calculate the potential difference across the 10 resistor. (3)
- 4.5 When switch S is opened the reading on voltmeter V_1 changes to **45,9 V**.
Hence, calculate the internal resistance of the battery. (5)
- 4.6 Does the external resistance in the circuit **INCREASE, DECREASE** or
REMAIN THE SAME when the resistor R is removed? (1)

MARATHON PROBLEM 6

In the circuit represented below, the lamp L has a resistance of $43\ \Omega$, and the internal resistance of the battery is $5\ \Omega$. The reading on the voltmeter **decreases** by $4,5\ \text{V}$ when switch S is closed.

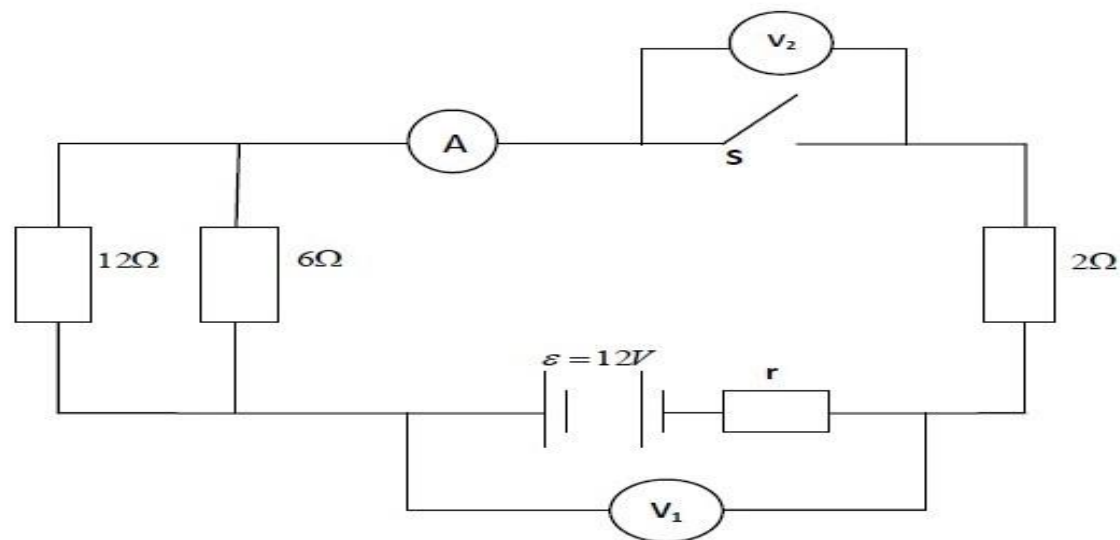


- 5.1 What causes this decrease in the voltmeter reading? (2)
- 5.2 Calculate:
- 5.2.1 the current in the battery when switch S is closed. (4)
- 5.2.2 the emf of the battery. (7)
- 5.2.3 the reading on the ammeter when switch S is closed. (5)
- 5.3 Will the power dissipated by the lamp **INCREASE**, **DECREASE** or **REMAIN CONSTANT** should resistor R_2 burn out while the switch is closed? Give a reason for the answer. (3)

MARATHON PROBLEM 7

(2)

The battery in a circuit diagram below has an EMF of 12 V and an unknown internal resistance r . **Voltmeter V_1** is connected across the battery and **Voltmeter V_2** is connected across the **switch S** . The resistance of the connecting wires and the ammeter is negligible.



7.1.1 Write down the respective readings on voltmeters V_1 and V_2 when the switch S is open.

(2)

Switch S is now closed. The reading on voltmeter V_1 changes to 9V.

7.1.2. What will the new reading on V_2 be?

(1)

7.1.3. Calculate the total external resistance of the circuit.

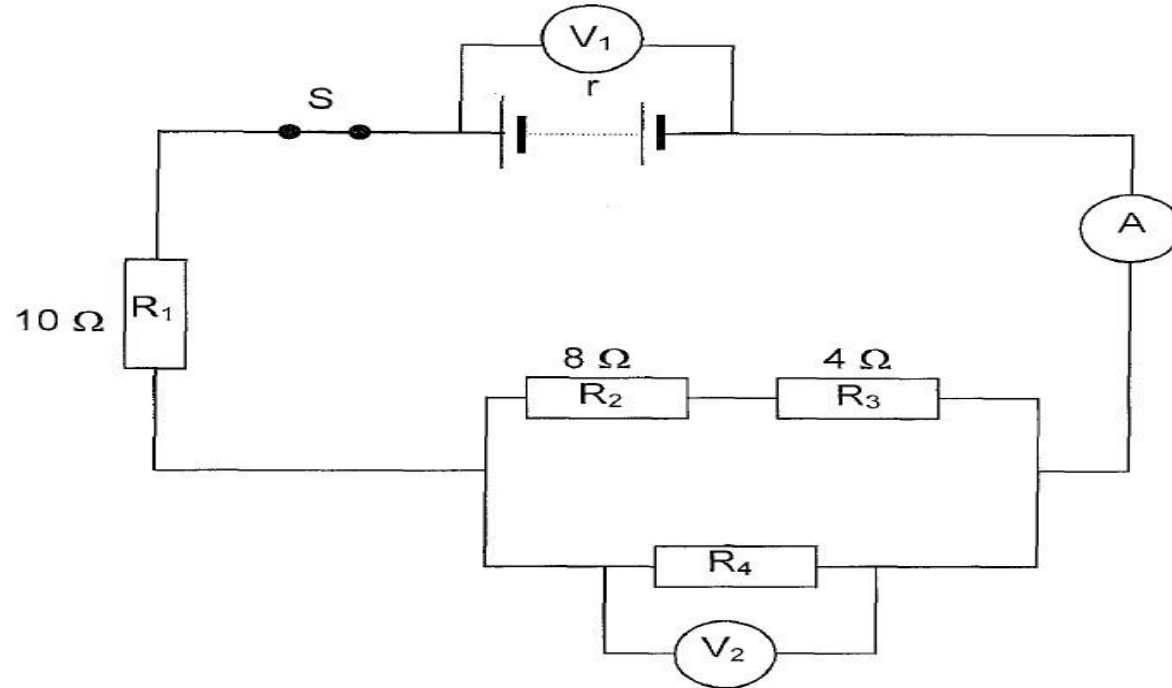
(4)

7.1.4. Calculate the internal resistance, r of the battery.

(5)

MARATHON PROBLEM 8

The battery in the circuit represented below, has an internal resistance r . The switch, S , is initially closed. The power dissipation in resistor R_4 is $13,5 \text{ W}$, while the reading on voltmeter V_2 is 18 V .



Calculate:

- 6.1 The resistance of R_4 (4)
- 6.2 The reading on the ammeter (6)

When switch S is opened the reading on voltmeter V_1 changes to $45,9 \text{ V}$.

- 6.3 Calculate the internal resistance, r , of the battery. (8)